

Characterization of Gaussian Universality Breakdown in High-Dimensional Empirical Risk Minimization

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Abstract

We study high-dimensional convex empirical risk minimization (ERM) under general non-Gaussian data designs. By heuristically extending the Convex Gaussian Min–Max Theorem (CGMT) to non-Gaussian settings, we derive an asymptotic min–max characterization of key statistics, enabling approximation of the mean $\mu_{\hat{\theta}}$ and covariance $C_{\hat{\theta}}$ of the ERM estimator $\hat{\theta}$. Specifically, under a concentration assumption on the data matrix and standard regularity conditions on the loss and regularizer, we show that for a test covariate x independent of the training data, the projection $\hat{\theta}^\top x$ approximately follows the convolution of the (generally non-Gaussian) distribution of $\mu_{\hat{\theta}}^\top x$ with an independent centered Gaussian variable of variance $\text{tr}(C_{\hat{\theta}} \mathbb{E}[xx^\top])$. This result clarifies the scope and limits of Gaussian universality for ERMs. Numerical simulations across diverse losses and models are provided to validate our theoretical predictions and qualitative insights.

1. Introduction

Given a training set of n samples $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^p \times \mathcal{Y}$, we consider convex regularized empirical risk minimization (ERM) problems of the form

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^p} \left\{ \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{y_i}(x_i^\top \theta) + \rho(\theta) \right\}, \quad (1)$$

where $\rho: \mathbb{R}^p \rightarrow \mathbb{R}$ is a convex regularizer and, for each $y \in \mathcal{Y}$, $\mathcal{L}_y: \mathbb{R} \rightarrow \mathbb{R}$ is a convex loss function. We focus on the high-dimensional regime in which both the sample size

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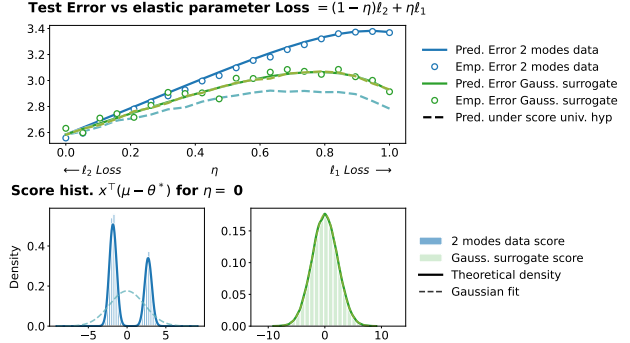


Figure 1. Breakdown of performance and score universality through an elastic loss. *Top:* Empirical generalization error and theoretical prediction for the teacher–student model $y = \theta^{*\top} x + \varepsilon$ as $\eta \in [0, 1]$ varies, with $\mathcal{L}_y(t) = (1 - \eta)(t - y)^2 + \eta|t - y|$. The entries of x are independent and bimodal, $\rho(\theta) = \|\theta\|_2^2$, $p = 30$, and $n = 50$. Performance universality is tested against Gaussian surrogate data with matching mean and covariance. *Bottom:* Score histograms at $\eta = 0$, together with the theoretical prediction and a moment-matched Gaussian fit. Performance universality holds at the squared-loss endpoint $\eta = 0$ although the score is non-Gaussian; for $\eta > 0$, the moment-matched Gaussian surrogate no longer predicts the empirical performance.

n and the dimension p are large, with typically $p = \Theta(n)$. As the minimizer of a random objective in (1), the solution $\hat{\theta}$ is itself random, through its dependence on the training set.

A central object for prediction is the *test score* $x^\top \hat{\theta}$, evaluated at a fresh covariate vector x independent of the training data. Many performance metrics for regression and classification (e.g., prediction or classification error) can be expressed as functions of (the distribution of) $x^\top \hat{\theta}$. The goal of the present paper is to develop a general framework to characterize its asymptotic law, and hence to predict the performance of ERM, in high dimensions, under *general* (possibly *non-Gaussian*) data and convex losses.

Several complementary theoretical toolkits have been developed to yield sharp high-dimensional predictions for ERM under various random designs. Random matrix theory (RMT) provides a natural framework when $p = \Theta(n)$. Starting with the seminal work of Karoui et al. (2013), RMT

yields precise deterministic equivalents for statistics of $\hat{\theta}$:

$$\mu_{\hat{\theta}} := \mathbb{E}[\hat{\theta}] \quad \text{and} \quad C_{\hat{\theta}} := \mathbb{E}[\hat{\theta}\hat{\theta}^\top] - \mu_{\hat{\theta}}\mu_{\hat{\theta}}^\top. \quad (2)$$

in specific models, often through implicit fixed-point characterizations (Bean et al., 2013; Karoui, 2018; Couillet et al., 2018; Louart et al., 2018; Seddik et al., 2020; Couillet & Liao, 2022; Dobriban & Wager, 2018). Approximate message passing (AMP) and its associated state evolution offer another route to exact asymptotic formulas for the risk and distribution of high-dimensional M -estimators (Bayati & Montanari, 2011; Rangan, 2011; Donoho & Montanari, 2016; Loureiro et al., 2021b; Han & Shen, 2022; Dandi et al., 2023; Montanari & Saeed, 2022; Pesce et al., 2023). Finally, the convex Gaussian min–max theorem (CGMT) reduces certain high-dimensional convex optimization problems with Gaussian design matrices to lower-dimensional auxiliary optimization (AO) problems, whose optimal values (and in many cases optimizers) closely track those of the original program (Gordon, 1988; Thrampoulidis et al., 2014; 2015; 2018; Loureiro et al., 2021a; Akhtiamov et al., 2024; Mallory et al., 2025; Bosch & Panahi, 2025).

Although developed from different starting points, these approaches ultimately lead to essentially the *same* system of fixed-point equations under Gaussian design. Moreover, the fixed-point equations arising in RMT and AMP can often be interpreted as reformulations of the KKT conditions of the CGMT min–max/AO problem—a viewpoint that will also play a role in this paper. What differs across methods is therefore primarily the class of models under which the corresponding arguments can be rigorously justified.

Despite these differences, all three approaches typically encounter a recurring obstacle. Either the analysis is restricted to the special case of Gaussian design, or, when extending to non-Gaussian settings, one ultimately relies on some form of *Gaussian universality* argument. Such arguments seek to “replace” the data or the test score by a Gaussian proxy with matching first- and second-order moments. In this paper, we provide a novel analysis framework that makes this universality step *explicit*, and allows us to precisely characterize when it holds and when it breaks down. Figure 1 illustrates this distinction in a teacher–student regression model with bimodal features: at the squared-loss endpoint, performance universality holds despite a non-Gaussian score, whereas away from that endpoint the Gaussian-surrogate performance prediction fails. Appendix G provides a related analysis on MNIST.

1.1. Main Contributions

Our starting point is a conjectural extension of the CGMT to a broad class of non-Gaussian designs; conditional on this comparison principle, we derive consequences that recover and generalize a large fraction of the exact-asymptotics

literature for smooth losses. Our main contributions are as follows:

1. **(Theorem 3.3)** Within the conditional framework below, we show that a smooth strongly convex regularizer satisfying weak-separability assumptions (Assumption 3) can be replaced, for the first two asymptotic moments of an independent test score, by the quadratic surrogate

$$\rho_q(\theta) := (\nabla \rho(\mu_{\hat{\theta}}) - H_\rho \mu_{\hat{\theta}})^\top \theta + \frac{1}{2} \theta^\top H_\rho \theta, \quad (3)$$

where $H_\rho := \nabla^2 \rho(0)$.

2. **(Theorem 4.3)** Conditional on Claim 4.1 and the saddle-point stability conditions stated below, we characterize the ERM asymptotics through the deterministic objective $\mathcal{J}$, defined, for $\mu \in \mathbb{R}^p$ and $\alpha, \nu, \beta, \kappa > 0$, by:

$$\begin{aligned} \mathcal{J}(\mu, \alpha, \kappa; \beta, \nu) &= \frac{\beta^2 \kappa}{2} + \mathbb{E} [e_{\mathcal{L}_y}(\mu^\top x + \alpha z; \kappa)] + \rho(\mu) \\ &\quad - \frac{\nu \alpha^2}{2} - \frac{\beta^2}{2n} \text{tr}(C_x(\nu C_x + H_\rho)^{-1}), \end{aligned} \quad (4)$$

where C_x is the covariance of x , $z \sim \mathcal{N}(0, 1)$ is independent of (x, y) , and $e_{\mathcal{L}_y} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ denotes the Moreau envelope defined, for any $u \in \mathbb{R}$, $\kappa > 0$ as:

$$e_{\mathcal{L}_y}(u; \kappa) := \min_{v \in \mathbb{R}} \left\{ \frac{1}{2\kappa} (u - v)^2 + \mathcal{L}_y(v) \right\}. \quad (5)$$

Let the saddle variables be defined by

$$\begin{aligned} (\mu_*, \alpha_*, \kappa_*) &\in \arg \min_{\mu \in \mathbb{R}^p, \alpha, \kappa > 0} \max_{\beta, \nu > 0} \mathcal{J}(\mu, \alpha, \kappa; \beta, \nu), \quad (6) \\ (\beta_*, \nu_*) &\in \arg \max_{\beta, \nu > 0} \mathcal{J}(\mu_*, \alpha_*, \kappa_*; \beta, \nu). \end{aligned}$$

Theorem 4.3 conditionally identifies (μ_*, α_*) with $(\mu_{\hat{\theta}}, \sqrt{\text{tr}(C_{\hat{\theta}} C_x)})$.

3. **(Theorem 6.1)** Under the estimator-level assumption that $\hat{\theta}$ behaves asymptotically as a Gaussian vector (Assumption 4), bounded Lipschitz observables of the test score are asymptotically described by

$$x^\top \hat{\theta} \approx x^\top \mu_* + \alpha_* z, \quad z \sim \mathcal{N}(0, 1),$$

with z independent of (x, y) . At the level of limiting laws, the score is Gaussian if and only if the informative projection $x^\top \mu_*$ is Gaussian.

4. **(Theorem 7.1)** For signal-plus-Gaussian-noise models $x = x_I + x_N$, where $x_I \in F$ almost surely and $x_N \sim \mathcal{N}(0, C_N)$ is independent of (x_I, y) and for quadratic regularizer $\rho : \theta \mapsto \frac{1}{2} \theta^\top H \theta$, we establish the covariance-aware confinement relation

$$\mu_* \in (H + \nu_* C_N)^{-1} F$$

Table 1. Comparison of related works and our method. Color code: **Light Red** = RMT, **Light Green** = CGMT, **Light Blue** = AMP, **Light Purple** = Lindeberg-type arguments (and variants). **Notation:** $\checkmark$ indicates that the corresponding work covers the feature/setting; **R** denotes a proved statement; **A** denotes an assumption. Parenthesized (**R**) indicates that the statement holds only under a subset of the assumptions listed in that column. Note that **R** entries may rely on different baseline assumptions across columns and are therefore not always directly comparable. Papers are grouped within a single column when they share a closely related framework and essentially the same conclusions, although some modeling assumptions may differ.

	(KOR11)	(ELK13) (BEA13)	(STO13) (THR18)	(DON16)	(PAN17) (HAN23)	(DOB18)	(ELK18)	(COU18) (MAI20)	(LOU21a) (LOU21b)	(MON22) (DAN23)	(HU22)	(PES23)	(ADO24)	(MAI25)	(AKH24) (BOS25)	(MAL25)	Our Method
Assumptions	Statistics of x																
	$\mu_x \neq 0$ (non-zero mean)					$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
	$C_x \neq \sigma^2 I_p$ (correlated features)					$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
	Multiple statistics							$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
	Law type, default: $x \sim \mathcal{N}(\mu_x, C_x)$																
	iid non-Gaussian entries	$\checkmark$			$\checkmark$	$\checkmark$	$\checkmark$			$\checkmark$	$\checkmark$			$\checkmark$		$\checkmark$	$\checkmark$
	Elliptical / linear factor						$\checkmark$			$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$			$\checkmark$
	Random features									$\checkmark$	$\checkmark$					$\checkmark$	$\checkmark$
	Heavy-tailed component			$\checkmark$	$\checkmark$								$\checkmark$				$\checkmark$
	Dependence $x \leftrightarrow y$																
	$y = \theta^{*\top} x + \varepsilon$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$
	y is class label of x					$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
	$y = f(\theta^{*\top} x)$									$\checkmark$	$\checkmark$	$\checkmark$				$\checkmark$	$\checkmark$
	Loss (default: ℓ_2)																
	General convex on score		$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$
	Regularization, default: None																
	Ridge (ℓ_2)			$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
	Lasso (ℓ_1)	$\checkmark$		$\checkmark$	$\checkmark$				$\checkmark$			$\checkmark$			$\checkmark$		$\checkmark$
	General convex			$\checkmark$	$\checkmark$				$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$
Results	Minimizer universality		R		R		R										A
	Score universality		R							A						A	(R)
	Performance universality	R	R		R	R	R	R		R	(R)	(R)	(R)	(R)		R	(R)
	Non-universality breakdown										R	R	R	R			R
	Deterministic perf. prediction		R	R	R		R	R	R		R	R	R	R	R		R

1.2. Related Work on Gaussian Universality

Since the literature uses the term “universality” in several inequivalent ways, we explicitly distinguish three distinct types of *Gaussian universality*:

1. *performance universality* holds if key performance metrics such as prediction risk or classification error remain asymptotically the same as if the design vectors $(x_1, \dots, x_n)$ were replaced by Gaussian covariates with matching mean and covariance.
2. *score universality* holds if, for an independent test vector x , the scalar score $x^\top \hat{\theta}$ converges in distribution to a Gaussian random variable; and
3. *minimizer universality* holds if the ERM solution $\hat{\theta}$ is asymptotically Gaussian.

Over the past decade, these questions have been investigated across a variety of settings; we summarize the resulting landscape in Table 1. For completeness, we additionally include two supplementary categories, distinguishing (i) works that exhibits explicit breakdowns of universality, and (ii) works that derive performance predictions based

on first-order statistics. Indeed, a number of previous efforts (Korada & Montanari, 2011; Panahi & Hassibi, 2017; Han & Shen, 2022; Montanari & Saeed, 2022; Dandi et al., 2023; Mallory et al., 2025) establish performance universality by leveraging Gaussian design surrogates to characterize asymptotic behavior under non-Gaussian data. These results typically aim to establish equivalence Gaussian and non-Gaussian models, rather than to provide explicit deterministic formulas for performance in the high-dimensional regime $p = \Theta(n)$ —which is the specific focus of the bottom row of Table 1.

Among the three notions, *performance universality* has the most direct practical implications and has been studied extensively, starting with the use of Lindeberg method (Chatterjee, 2006) in Korada & Montanari (2011); Panahi & Hassibi (2017); Han & Shen (2022); Goldt et al. (2022). Three primary categories of assumptions/settings are known to yield performance universality:

1. **Gaussian covariates:** when x is Gaussian, precise characterizations are available (Karoui et al., 2013; Bean et al., 2013; Karoui, 2018; Couillet et al., 2018; Mai & Liao, 2020). Note, however, that Gaussian *mixtures* may invalidate these conclusions, see Pesce et al. (2023).

2. **Squared ℓ_2 loss:** in the special case of squared loss, many *performance universality* results have been obtained (Korada & Montanari, 2011; Panahi & Hassibi, 2017; Han & Shen, 2022). In Section 5, we demonstrate that for squared loss, performance universality can hold *even when score universality fails*.
3. **Asymptotic score Gaussianity:** Recent works (Montanari & Saeed, 2022; Mallory et al., 2025) derive performance universality by assuming *score universality*, a condition we discuss below in greater detail.

To establish performance universality, Montanari & Saeed (2022) and subsequently Mallory et al. (2025) rely on assumptions that, while seemingly mild, effectively enforce a central limit theorem for $x^\top \hat{\theta}$ (identified in Table 1 as *score universality*). Specifically, they assume the existence of a (“unavoidable”) subset $S \subset \mathbb{S}^{p-1}$ such that for any bounded Lipschitz function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ and any Gaussian vector g matching the first two moments of x ,

$$\sup_{u \in S} \left| \mathbb{E}[\phi(u^\top x)] - \mathbb{E}[\phi(u^\top g)] \right| \xrightarrow{n \rightarrow \infty} 0.$$

If one can further guarantee that $\hat{\theta}/\|\hat{\theta}\| \in S$ with high probability, it follows that

$$\mathbb{E}[\phi(\hat{\theta}^\top x)] - \mathbb{E}[\phi(\hat{\theta}^\top g)] \xrightarrow{n \rightarrow \infty} 0. \quad (7)$$

However, this condition constitutes a significant limitation. It has been observed in several settings (Hu & Lu, 2022; Pesce et al., 2023; Mai & Liao, 2025) that the direction of the estimator may align with specific structural features of the data, resulting in $\mu_{\hat{\theta}}/\|\mu_{\hat{\theta}}\| \notin S$. In such cases, score universality—and consequently the universality of performance metrics derived via this route—breaks down. A central contribution of this paper is to characterize, as broadly as possible, the conditions under which such *Gaussian universality breakdown* occurs (Figure 1 and Figure 2 in the appendix show two examples where both score and performance universality are invalidated).

Finally, *minimizer universality* has received comparatively less attention, despite being plausibly generic for smooth $\mathcal{L}_{y_i}$ and ρ . The first-order optimality condition for (1) is

$$0 = \frac{1}{n} \sum_{i=1}^n \mathcal{L}'_{y_i}(x_i^\top \hat{\theta}) x_i + \nabla \rho(\hat{\theta}).$$

Using the quadratic approximation of ρ from (3), this yields:

$$H_\rho(\hat{\theta} - \mu_{\hat{\theta}}) \approx -\nabla \rho(\mu_{\hat{\theta}}) - \frac{1}{n} \sum_{i=1}^n \mathcal{L}'_{y_i}(x_i^\top \hat{\theta}) x_i. \quad (8)$$

This suggests that $\hat{\theta}$ behaves as an average of many (weakly dependent) random vectors, and may therefore satisfy a

Lindeberg-type central limit theorem. Motivated by this heuristic, we introduce minimizer universality as a standing assumption (Assumption 4) to establish full characterization of the test score. Moreover, it must be noted that this assumption is *not* required for the *non-Gaussian* CGMT-based pipeline in our Theorem 4.3.

1.3. Notations and Organization of the Paper

Throughout the paper, we adopt the notations introduced in (1). We denote the data matrices by $X = (x_1, \dots, x_n) \in \mathbb{R}^{p \times n}$ and $Y = (y_1, \dots, y_n) \in \mathcal{Y}^n$. A test sample independent of the training data (X, Y) (and thus of $\hat{\theta}$) is denoted by (x, y) , and $z \sim \mathcal{N}(0, 1)$ denotes a standard Gaussian variable independent of all data. We write $\mu_x := \mathbb{E}[x]$, $C_x := \text{Cov}(x)$, and $\Sigma_x := \mathbb{E}[xx^\top] = C_x + \mu_x \mu_x^\top$.

The Euclidean norm of a vector in $\mathbb{R}^p$ is denoted by $\|\cdot\|$. For matrices and tensors, $\|\cdot\|$ denotes the ℓ_2 operator norm (spectral norm); specifically, for a tensor $T \in \mathbb{R}^{d_1 \times \dots \times d_k}$, we define $\|T\| := \sup_{\|h_i\|=1} |T[(h_i)_{i \in [k]}]|$. For a matrix $M \in \mathbb{R}^{p \times n}$, we denote the Frobenius norm by $\|M\|_F = \sqrt{\text{tr}(MM^\top)}$ and the nuclear norm by $\|M\|_* = \text{tr}((MM^\top)^{\frac{1}{2}})$.

The remainder of this paper is organized as follows. Section 2 introduces the concentration and regularity assumptions used throughout the paper. Section 3 shows that, under suitable smoothness assumptions, the regularizer can be asymptotically replaced by a quadratic surrogate depending only on its local second-order structure. Section 4 presents a conditional extension of the CGMT framework to concentrated non-Gaussian designs and derives the corresponding deterministic min-max and fixed-point characterizations for the asymptotic statistics of the ERM estimator. Section 5 illustrates the framework on ridge regression under a linear model, in which case performance universality can hold even when score universality fails. Section 6 studies the asymptotic law of the test score and characterizes precisely when Gaussian score universality holds or breaks down. Finally, Section 7 provides covariance-adjusted confinement results for μ_* in structured signal-plus-Gaussian-noise models and identifies the structured source of possible score non-Gaussianity.

2. Preliminaries and Problem Setup

We assume that the training set consists of n independent samples $(x_1, y_1), \dots, (x_n, y_n)$ drawn from a common distribution on $\mathbb{R}^p \times \mathcal{Y}$.¹ To obtain finite- n bounds, we adopt assumptions inspired by concentration-of-measure theory. As shown in (Seddik et al., 2020), such assumptions encompass

¹Our results naturally extend to multiclass settings; see Section A in the appendix.

a broad class of realistic designs, including those generated via Lipschitz transformations of Gaussian latent variables, as commonly encountered in generative models. We work in the proportional regime $0 < c \leq p/n \leq C < \infty$. Throughout, we track the explicit dependence of our bounds on n , while absorbing quantities that do not affect this scaling into generic constants $C, c > 0$.

We equip one sample with

$$d_0((x, y), (x', y'))^2 := \|x - x'\|^2 + d_{\mathcal{Y}}(y, y')^2,$$

and use the associated ℓ_2 -product metric for several samples. Let $Z := \text{vec}(x, y)$ denote a fixed vectorization; D denotes the corresponding Euclidean gradient (or the natural difference gradient for categorical coordinates). We impose a Poincaré inequality at the one-sample level.²

Assumption 1 (Samplewise Poincaré concentration). *There exist constants $c, C, C_P > 0$, independent of n, p , such that $cn \leq p \leq Cn$, $\|\mathbb{E}[x]\| \leq C$, and, for every sufficiently regular φ ,*

$$\text{Var}(\varphi(Z)) \leq C_P \mathbb{E}[\|D\varphi(Z)\|^2]. \quad (9)$$

For $Z_0 := \text{vec}(x, y)$ and $Z_i := \text{vec}(x_i, y_i)$, independence and tensorization yield

$$\text{Var}(F(Z_0, \dots, Z_n)) \leq C_P \sum_{i=0}^n \mathbb{E}[\|D_i F(Z_0, \dots, Z_n)\|^2].$$

Thus arbitrary dependence is allowed within each pair (x_i, y_i) ; only the sample blocks are independent.

Assumption 1 holds for nonlinear regression model with data $x_i = \Phi(z_i)$ for $z_i \sim \mathcal{N}(0, I_p)$ and Lipschitz Φ , and target $y_i = g(x_i, w_i)$ being a Lipschitz transformation (g) of x_i and Gaussian noise w_i .³ Similarly, binary class-conditional models $x_i = \phi_{y_i}(z_i)$ satisfy Assumption 1 whenever the latent pair (y_i, z_i) obeys a dimension-free product Poincaré inequality and the maps ϕ_1, ϕ_2 are uniformly Lipschitz.

We next impose regularity conditions on the loss and regularizer to ensure (i) the existence and uniqueness of the ERM solution $\hat{\theta}$ and (ii) that the concentration properties of the data in Assumption 1 transfers smoothly to $\hat{\theta}$.

Assumption 2 (Regularity of loss and regularizer). *For all $y \in \mathcal{Y}$, the functions $\mathcal{L}_y : \mathbb{R} \rightarrow \mathbb{R}$ and $\rho : \mathbb{R}^p \rightarrow \mathbb{R}$ are convex and twice continuously differentiable. We*

²The submitted version assumed concentration directly for the full sample. Following the subsequent Wasserstein CLT for regularized M -estimators (Louart, 2026), we instead impose the analogous columnwise Poincaré hypothesis on (x_i, y_i) ; tensorization gives the full training-and-test statement.

³In this setting, by standard Gaussian concentration results (Ledoux, 2005), Equation (9) holds (one can actually state a stronger concentration inequality).

assume that $\mathcal{L}_y(0)$ and $\|\nabla \rho(0)\|$ are bounded uniformly in n . Furthermore:

1. *There exists $\lambda > 0$ such that $\nabla \rho$ is λ -Lipschitz, and for any $y \in \mathcal{Y}$, the derivative $\mathcal{L}'_y$ is λ -Lipschitz. Additionally, the map $y \mapsto \mathcal{L}'_y(u)$ is λ -Lipschitz (with respect to $d_{\mathcal{Y}}$) for any fixed u .*
2. *There exists $\kappa > 0$, independent of p and n , such that $\nabla^2 \rho(\theta) \succeq \kappa I_p$ for all $\theta \in \mathbb{R}^p$.*

Assumption 2 excludes non-smooth penalties such as the Lasso. While the Lasso can be analyzed via CGMT (Stojnic, 2013), the strict convexity and smoothness assumed here are required to ensure that $\hat{\theta}$ concentrates strongly and satisfies the weakly dependent average formulation (8).

These two assumptions suffice to establish the concentration of the minimizer $\hat{\theta}$, which is fundamental to our subsequent analysis of the score's asymptotic behavior.

Theorem 2.1 (Concentration of $\hat{\theta}$). *Under Assumptions 1 and 2, for every fixed $q \geq 2$, there is a constant $C_q > 0$, independent of n, p , such that, for every 1-Lipschitz $f : \mathbb{R}^p \rightarrow \mathbb{R}$,*

$$\|f(\hat{\theta}) - \mathbb{E}f(\hat{\theta})\|_{L^q} \leq \frac{C_q}{\sqrt{n}}.$$

In particular, $\text{Var}(f(\hat{\theta})) = O(n^{-1})$.

Theorem 2.1, combined with the concentration property of x (Assumption 1), provides bounds on the operator norm of the covariance matrices C_x and $C_{\hat{\theta}}$ of x and $\hat{\theta}$. This result will be further explored to establish the asymptotic behavior of test scores in Theorem 6.1.

Corollary 2.2 (Louart (2024)). *Under Assumptions 1 and 2, we have $\|C_x\| = O(1)$ and $\|C_{\hat{\theta}}\| = O(1/n)$.*

3. Quadratic Universality for Smooth Regularizers

We now demonstrate that, on the scale relevant for high-dimensional fluctuations, the regularizer can be effectively approximated by a quadratic form derived from a second-order Taylor expansion around $\mu_{\hat{\theta}}$. To control the approximation error, we impose bounds on the operator norm of the third-derivative tensor $\nabla^3 \rho$. When ρ is separable, $\nabla^3 \rho(\theta)$ is a diagonal tensor for all $\theta \in \mathbb{R}^p$, and it is natural to assume that the diagonal entries do not grow with the dimension p (as is the case for the ℓ_2 penalty). For non-separable regularizers, the Taylor approximation requires additional control over off-diagonal interactions.

Assumption 3. *The function $\rho : \mathbb{R}^p \rightarrow \mathbb{R}$ is $\mathcal{C}^3$ and there*

exists a constant $C > 0$ such that, for all $\theta \in \mathbb{R}^p$:⁴

$$\|\nabla^3 \rho(\theta)\| \leq C \quad \text{and} \quad \|\text{OffDiag}(\nabla^3 \rho(\theta))\| \leq Cp^{-1/2}.$$

Note that the condition on the off-diagonal tensor entries is weaker than the “weakly separable” hypothesis typically assumed in the literature; see, e.g., (Panahi & Hassibi, 2017). A second-order Taylor expansion of ρ around $\mu_{\hat{\theta}}$ yields the following concentration bound for the remainder.

Proposition 3.1. *Under Assumptions 1, 2, and 3, let*

$$R_n := \left| \rho(\hat{\theta}) - \rho(\mu_{\hat{\theta}}) - \nabla \rho(\mu_{\hat{\theta}})^\top (\hat{\theta} - \mu_{\hat{\theta}}) - \frac{1}{2} (\hat{\theta} - \mu_{\hat{\theta}})^\top \nabla^2 \rho(\mu_{\hat{\theta}}) (\hat{\theta} - \mu_{\hat{\theta}}) \right|.$$

Then

$$\mathbb{E}[R_n] = O(n^{-1/2}),$$

and in particular $R_n \xrightarrow{\mathbb{P}} 0$.

We saw in (4) that $\nabla^2 \rho$ enters our asymptotic formulas primarily through the trace term:

$$\frac{1}{n} \text{tr}(C_x Q_\mu), \quad \text{where} \quad Q_\mu := (\nabla^2 \rho(\mu) + \nu C_x)^{-1}.$$

In this expression, one may replace $\nabla^2 \rho(\mu)$ with its value at zero,

$$H_\rho := \nabla^2 \rho(0),$$

without affecting the asymptotic limit, thanks to the following lemma.

Lemma 3.2. *Under Assumption 3, there exists a constant $C > 0$, independent of n and p , such that for all $\theta \in \mathbb{R}^p$, $\|H_\rho - \nabla^2 \rho(\theta)\|_F \leq C\|\theta\|$. Consequently, there exists C' depending on C such that:*

$$\frac{1}{n} \text{tr}(C_x(Q_\mu - Q_0)) \leq \frac{C'}{\sqrt{n}}, \quad \text{with } Q_0 := (H_\rho + \nu C_x)^{-1}.$$

Let $\hat{\theta}_q$ solve the same ERM problem as $\hat{\theta}$ with ρ replaced by

$$\rho_q(\theta) := a^\top \theta + \frac{1}{2} \theta^\top H_\rho \theta, \quad a := \nabla \rho(\mu_{\hat{\theta}}) - H_\rho \mu_{\hat{\theta}}.$$

Thus $\nabla \rho_q(\mu_{\hat{\theta}}) = \nabla \rho(\mu_{\hat{\theta}})$ and $\nabla^2 \rho_q = H_\rho$.

Theorem 3.3 (Quadratic universality of regularization). *Under Assumptions 1, 2 and 3:*

$$\left| \mathbb{E}[x^\top \hat{\theta}] - \mathbb{E}[x^\top \hat{\theta}_q] \right|, \left| \mathbb{E}[(x^\top \hat{\theta})^2] - \mathbb{E}[(x^\top \hat{\theta}_q)^2] \right| \xrightarrow{n \rightarrow \infty} 0.$$

This result is a consequence of Proposition 3.1, Lemma 3.2 and Theorem 4.3 presented below.

⁴For a tensor T , $\|T\|$ denotes the operator norm $\sup_{\|u\|=1} |T[u, u, u]|$. For the off-diagonal part, we refer to the operator norm of the tensor with diagonal entries set to zero.

4. Non-Gaussian CGMT and Implications for ERM Asymptotics

The Convex Gaussian Min–Max Theorem (CGMT) transforms a min–max problem involving a random Gaussian matrix $A \in \mathbb{R}^{p \times n}$ into an associated *auxiliary problem* whose randomness depends only on two Gaussian vectors $g \in \mathbb{R}^p$ and $h \in \mathbb{R}^n$ (Gordon, 1988). Originally used to study the singular values of Gaussian matrices (Vershynin, 2012; Oymak & Tropp, 2018), this tool has become a cornerstone for analyzing high-dimensional risk minimization, beginning with (Stojnic, 2013; Thrampoulidis et al., 2014).

The first step is to rewrite the ERM problem (1) as a constrained optimization problem with an objective linear in the random design matrix X :

$$\min_{\theta \in \mathbb{R}^p, v \in \mathbb{R}^n} \left\{ \frac{1}{n} \mathcal{L}_y(v) + \rho(\theta) \right\} \quad \text{s.t.} \quad v = X^\top \theta.$$

Introducing a dual variable $w \in \mathbb{R}^n$ for the constraint yields the Lagrangian:

$$\mathcal{L}(\theta, v, w) = w^\top (X^\top \theta - v) + \frac{1}{n} \mathcal{L}_y(v) + \rho(\theta).$$

Minimization over v and maximization over w commute, leading to the following min–max formulation suited for an application of the CGMT:

$$\hat{\theta} \in \arg \min_{\theta \in \mathbb{R}^p} \max_{w \in \mathbb{R}^n} \left\{ \theta^\top X w - \frac{1}{n} \mathcal{L}_y^*(nw) + \rho(\theta) \right\}, \quad (10)$$

where $\mathcal{L}_y^*$ denotes the Fenchel conjugate of $\mathcal{L}_y$, i.e., $\mathcal{L}_y^*(w) = \sup_{v \in \mathbb{R}^n} \{v^\top w - \mathcal{L}_y(v)\}$.

In the claim below, we isolate the specific properties of the generalized CGMT required for our derivations (analogous to Thrampoulidis et al. (2014, Theorem 4)).

Claim 4.1 (Generalized CGMT for concentrated column designs). *Let $C_1 \subset \mathbb{R}^p$ and $C_2 \subset \mathbb{R}^n$ be convex compact sets, and let ψ_y be convex–concave. For $A \in \mathbb{R}^{p \times n}$, $\mu, a \in \mathbb{R}^p$, and $h \in \mathbb{R}^n$, define*

$$\Phi(A) := \min_{\theta \in C_1} \mathcal{V}_\Phi(\theta, A),$$

$$\phi(a, h) := \min_{\theta \in C_1} \mathcal{V}_\Phi(\theta, a, h),$$

$$\mathcal{V}_\Phi(\theta, A) := \max_{w \in C_2} \{ \theta^\top A w + \mu^\top A w + \psi_y(\theta, w) \},$$

$$\mathcal{V}_\Phi(\theta, a, h) := \max_{w \in C_2} \{ w^\top h \|C_x^{1/2} \theta\| + \|w\| \theta^\top a + (\mu^\top a) \mathbf{1}^\top w + \psi_y(\theta, w) \}.$$

Let $\theta_\Phi(A)$ and $\theta_\phi(a, h)$ be the corresponding minimizers. Under Assumption 1, let $g \sim \mathcal{N}(0, I_n)$ be independent of (X, x) . We conjecture that, if

$$1. \quad \phi(x, g) - \mathbb{E}[\phi(x, g)] \xrightarrow{\mathbb{P}} 0;$$

$$2. \|\theta_\phi(x, g)\| - \mathbb{E}\|\theta_\phi(x, g)\| \xrightarrow{\mathbb{P}} 0;$$

3. for some $\tau > 0$ and all sufficiently large n ,

$$\mathcal{V}_\phi(\theta, x, g) \geq \phi(x, g) + \tau(\|\theta\| - \|\theta_\phi(x, g)\|)^2, \quad \forall \theta \in C_1,$$

then, for every $\varepsilon > 0$,

$$\begin{aligned} \mathbb{P}(|\Phi(X) - \phi(x, g)| > \varepsilon) &\rightarrow 0, \\ \mathbb{P}(\|\Phi(X) - \mathbb{E}\|\theta_\phi(x, g)\| > \varepsilon) &\rightarrow 0. \end{aligned}$$

Remark 4.2. Recent CGMT results such as those in (Akhtiamov et al., 2024; Bosch & Panahi, 2025), while treating multiclass settings, do not explicitly model the dependence of ψ on the label y , nor do they incorporate the deterministic shift parameter $\mu \in \mathbb{R}^p$. This distinction is not merely anecdotal; a central component of our approach is to leverage the concentration of $\hat{\theta}$ via the decomposition:

$$\hat{\theta} = \mu_{\hat{\theta}} + \theta_0, \quad (11)$$

and to analyze the minimization with respect to the fluctuation θ_0 and the mean $\mu_{\hat{\theta}}$ separately. When $\mu_{\hat{\theta}}$ is fixed, we apply our claim with $\mu = C_x^{\frac{1}{2}}\mu_{\hat{\theta}}$, $\psi_y(\theta_0, w) = -\frac{1}{n}\mathcal{L}_y^*(nw) + \rho(\mu_{\hat{\theta}} + C_x^{-\frac{1}{2}}\theta_0)$, and replace the random matrix X with the whitened version $C_x^{-\frac{1}{2}}X$. This yields theoretical predictions for quantities such as $\|C_x^{1/2}\theta_0\|$, which in our setting converges to $\sqrt{\text{tr}(C_x C_{\hat{\theta}})}$. In contrast, standard CGMT approaches would focus on estimating the total norm $\|C_x^{1/2}(\theta_0 + \mu_{\hat{\theta}})\|$, which is less amenable to the subsequent convex analysis machinery we employ.

After restricting the primal and dual variables to high-probability compact sets, Claim 4.1 yields the auxiliary problem used below. The resulting identification of the ERM with its deterministic saddle point remains conditional on this conjectural comparison principle and on the stability hypotheses stated next.

Theorem 4.3 (Min–max formulation of limiting asymptotics). *Under Assumptions 1, 2, and 3:*

$$\|\mu_{\hat{\theta}} - \mu_*\| \xrightarrow{n \rightarrow \infty} 0 \quad \text{and} \quad \text{tr}(C_x C_{\hat{\theta}}) - \alpha_*^2 \xrightarrow{n \rightarrow \infty} 0,$$

where (μ_*, α_*) are the saddle-point solutions to the min–max problem in (6).

In plain words, Theorem 4.3 states that the high-dimensional ERM estimator can be “summarized” asymptotically by two deterministic quantities: $\mu_* \approx \mathbb{E}[\hat{\theta}]$ that captures the deterministic informative component of the estimator and $\alpha_*^2 \approx \text{tr}(C_x C_{\hat{\theta}})$ that measures the scale of its random fluctuations. These quantities can be explicitly obtained from a deterministic saddle-point problem depending only on the data model (x, y) , the loss, and the local quadratic structure of

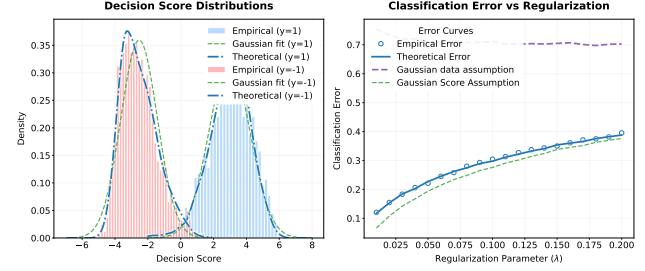


Figure 2. Universality breakdown on MNIST data. Setting described in Appendix G. *Left:* Empirical histograms of decision scores for Class 0 (light blue) and Class 1 (light red) of $\hat{\theta}^\top x$, compared with a Gaussian approximation of matching mean and variance (green dashed) and with the corrected theoretical density (dashed blue). *Right:* Generalization performance. As expected, predictions based on Gaussian score universality (green dashed) fail to match empirical results.

the regularizer. In this sense, the random high-dimensional optimization problem (1) admits a low-dimensional deterministic characterization.

Combined with Assumption 4, this conditional characterization gives the score proxy $x^\top \mu_* + \alpha_* z$. The corresponding Gaussianity criterion is stated precisely in Section 6.

A key novelty relative to previous analyses is the explicit role of the vector μ_* , which captures the (generally non-zero) asymptotic mean of $\hat{\theta}$ for general losses and data distributions. Theorem 4.3 reveals that the distribution of x influences the asymptotics specifically through the Gaussian-convolved variable $\mu^\top x + \alpha z$ (see again Section 6).

For theoretical analysis and numerical computation, it is convenient to reformulate the min–max problem as a fixed-point system. For any $y \in \mathcal{Y}$, we define the mapping $\xi_y : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}$ by:

$$\xi_y(u, \kappa) := u - \text{prox}_{\kappa \mathcal{L}_y}(u), \quad (12)$$

where the proximal operator is given by $\text{prox}_{\kappa \mathcal{L}_y}(u) := \arg \min_{w \in \mathbb{R}} \{\kappa \mathcal{L}_y(w) + \frac{1}{2}(u - w)^2\}$. We further define the resolvent $Q(\nu) := (\nu C_x + H_\rho)^{-1}$ and the functional $A(\nu) := \frac{1}{n} \text{tr}(C_x Q(\nu) C_x Q(\nu))$.

Theorem 4.4 (Fixed-point equations). *Any interior saddle point $(\mu_*, \alpha_*, \kappa_*; \beta_*, \nu_*)$ of (4) satisfies*

$$\begin{aligned} \kappa_* &= \frac{1}{n} \text{tr}(C_x Q(\nu_*)), \\ \nu_* &= \frac{1}{\alpha_* \kappa_*} \mathbb{E}[z \xi_y(\mu_*^\top x + \alpha_* z, \kappa_*)], \\ \alpha_*^2 &= \frac{A(\nu_*)}{\kappa_*^2} \mathbb{E}[\xi_y(\mu_*^\top x + \alpha_* z, \kappa_*)^2], \\ 0 &= \nabla \rho(\mu_*) + \frac{1}{\kappa_*} \mathbb{E}[x \xi_y(\mu_*^\top x + \alpha_* z, \kappa_*)]. \end{aligned}$$

The expectations are over (x, y) and an independent $z \sim \mathcal{N}(0, 1)$.

In the specific case of squared loss with a linear model, Theorem 4.4 reduces to the two-equation system derived in Karoui et al. (2013); Bean et al. (2013); Donoho & Montanari (2016) (see Section 5). More broadly, these equations recover and extend the form of fixed-point characterizations previously established for regularized regression (Thrapoulidis et al., 2015) and high-dimensional classification in mixture models (Mai & Liao, 2020; 2025).

5. Performance Universality of Ridge Regression under Linear Model

In this section, we apply our framework to the specific case of squared loss with a linear data model:

$$\mathcal{L}_y(t) = \frac{1}{2}(t - y)^2, \quad \text{where } y = \theta^{*\top} x + \varepsilon. \quad (13)$$

Here, $\theta^* \in \mathbb{R}^p$ is deterministic and ε is independent of x with

$$\mathbb{E}[\varepsilon] = 0, \quad \sigma_\varepsilon^2 := \mathbb{E}[\varepsilon^2]. \quad (14)$$

Recall that $\Sigma_x = C_x + \mu_x \mu_x^\top$. This setting mirrors the one analyzed in Dobriban & Wager (2018), with the generalization that X need not be a linear transformation of a random matrix with independent entries. Following the notation of the previous section, the proximal operator for the squared loss admits a closed-form expression, leading to:

$$\xi_y(u, \kappa) = \frac{\kappa}{1 + \kappa}(u - y).$$

This greatly simplifies the fixed-point equations in Theorem 4.4. Specifically, the relation between ν and κ becomes explicit, yielding a single equation for ν_* :

$$\nu_* = \frac{1}{1 + \kappa_*} = \frac{1}{1 + \frac{1}{n} \text{tr}(C_x Q(\nu_*))}. \quad (15)$$

Note that this identity is indeed satisfied for the ℓ_2 norm on Figure 3. This is the classical fixed-point equation from Random Matrix Theory, known to possess a unique solution (Silverstein & Bai, 1995). Leveraging the quadratic universality established in Theorem 3.3, we adopt the quadratic formulation for ρ below to derive a structural result.

Corollary 5.1 (Generalization error of ridge regression). Consider (13) with

$$\rho(\theta) = a^\top \theta + \frac{1}{2} \theta^\top H \theta, \quad H \succ 0.$$

Under the hypotheses of Theorem 4.3, let ν_* solve (15). Then

$$\mathbb{E}[\mathcal{L}_y(x^\top \hat{\theta})] - \frac{1}{2} \frac{\sigma_\varepsilon^2 + \Delta(\nu_*)}{1 - \nu_*^2 A(\nu_*)} \longrightarrow 0,$$

Description variables through elastic loss

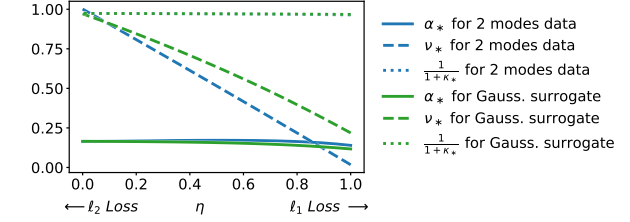


Figure 3. Evolution of the description variables through elastic loss. In the setting of Figure 1, we show the fixed-point solution variables α_* , ν_* and $\frac{1}{1+\kappa_*}$. For the ℓ_2 loss ($\eta = 0$), we have $\nu_* = \frac{1}{1+\kappa_*}$.

where

$$\begin{aligned} \mu(\nu) &:= (H + \nu \Sigma_x)^{-1}(\nu \Sigma_x \theta^* - a), \\ \Delta(\nu) &:= (\mu(\nu) - \theta^*)^\top \Sigma_x (\mu(\nu) - \theta^*). \end{aligned}$$

Thus the asymptotic generalization error depends on the law of x only through its mean μ_x and covariance C_x , confirming *performance universality*. Score universality may nevertheless fail, since $x^\top \hat{\theta}$ can remain non-Gaussian. This is exactly what happens when $\eta = 0$ on Figure 1.

6. Score Universality Characterization

Unlike the score universality condition (7) assumed in Montanari & Saeed (2022) and Mallory et al. (2025), we are not aware of any empirical counterexamples to the following assumption in the proportional regime. See again discussion around (8) to understand the underlying heuristic. A proof of this result based on Chatterjee’s normal-approximation method ((Chatterjee, 2008; Shao & Zhang, 2025)) is under preparation (see (Louart, 2026)).

Assumption 4 (Minimizer universality). For any bounded measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$ and any deterministic sequence $(u_n)_{n \in \mathbb{N}} \subset \mathbb{R}^p$ with $\|u_n\| = O(\sqrt{n})$:

$$\mathbb{E}[f(u_n^\top \hat{\theta})] - \mathbb{E}[f(u_n^\top g)] \xrightarrow{n \rightarrow \infty} 0,$$

where $g \sim \mathcal{N}(\mu_{\hat{\theta}}, C_{\hat{\theta}})$.

Uniformity over unit directions allows Assumption 4 to be applied conditionally to the independent direction $x/\|x\|$; see the appendix. Conditionally on x ,

$$x^\top g \mid x \sim \mathcal{N}(x^\top \mu_{\hat{\theta}}, x^\top C_{\hat{\theta}} x).$$

In fact, $x^\top C_{\hat{\theta}} x$ concentrates with high probability around $\text{tr}(\Sigma_x C_{\hat{\theta}})$ by Hanson–Wright inequality, which leads to the following result.

Theorem 6.1 (Asymptotic behavior of test score). Under Assumptions 1–4, for any bounded Lipschitz function $f :$

$\mathbb{R} \times \mathcal{Y} \rightarrow \mathbb{R}$:

$$\mathbb{E}[f(x^\top \hat{\theta}, y)] - \mathbb{E}[f(x^\top \mu_* + \alpha_* z, y)] \xrightarrow{n \rightarrow \infty} 0,$$

where (μ_*, α_*) are defined in Theorem 4.3 and $z \sim \mathcal{N}(0, 1)$ is independent of (x, y) .

Theorem 6.1 fully characterizes the score universality:

$$“x^\top \hat{\theta} \text{ Gaussian} \iff x^\top \mu_* \text{ Gaussian}”.$$

It is natural at this stage to relate our score characterization to the recent literature on Gaussian and conditional Gaussian equivalence for random feature models (Gerace et al., 2022; Dandi et al., 2023; Wen et al., 2025). A common formulation assumes the existence of a low-dimensional random vector $s \in \mathbb{R}^r$, with $r = O(1)$, such that

$$y = \eta(s, \varepsilon), \quad x | s \sim \mathcal{N}(m(s), \Sigma(s)),$$

where ε is independent of (x, s) . Under such a representation, our score formula yields

$$x^\top \hat{\theta} \approx m(s)^\top \mu_* + \sqrt{\mu_*^\top \Sigma(s) \mu_* + \alpha_*^2} z,$$

so that the fixed-point system of Theorem 4.4 reduces to expectations over the low-dimensional variable s . In general, both the random conditional mean $m(s)^\top \mu_*$ and the random conditional variance $\mu_*^\top \Sigma(s) \mu_*$ can generate a non-Gaussian mixture. Only when the latter is asymptotically constant is the non-Gaussian component carried solely by $m(s)^\top \mu_*$. We leave a systematic study of this reduced system to future work. We instead focus below on a structural consequence for signal-plus-Gaussian-noise models: the asymptotic mean is confined to a covariance-adjusted image of the structured subspace.

7. Covariance-Adjusted Subspace Confinement of $\mu_{\hat{\theta}}$

Although μ_* , the asymptotic limit of $\mu_{\hat{\theta}}$ in the conditional characterization, is determined by the min-max problem in (6), its direction remains implicit in general. For signal-plus-noise models, the vector fixed-point equation in Theorem 4.4 gives a more explicit geometric description. A key point is that anisotropic residual noise modifies the relevant geometry: the asymptotic mean is confined only after accounting for the covariance of the residual Gaussian component.

We consider decompositions of the form

$$x = x_I + x_N, \quad x_I \in F \text{ a.s.}, \quad x_N \sim \mathcal{N}(0, C_N), \quad (16)$$

where $F \subset \mathbb{R}^p$ is a subspace and x_N is independent of (x_I, y) . The structured component x_I may be non-Gaussian and carries all the dependence between the feature vector and the label.

Theorem 7.1 (Covariance-adjusted subspace confinement). *Under Assumptions 2 and 3, let $(\mu_*, \alpha_*, \kappa_*, \nu_*)$ satisfy Theorem 4.4, and assume (16). Then*

$$\nabla \rho(\mu_*) + \nu_* C_N \mu_* \in F.$$

Consequently, with

$$H_\rho := \nabla^2 \rho(0), \quad a_* := \nabla \rho(\mu_*) - H_\rho \mu_*,$$

one has

$$\mu_* \in (H_\rho + \nu_* C_N)^{-1} (F + \text{span}(a_*)). \quad (17)$$

The covariance correction in (17) is essential. For example, consider the Gaussian mixture

$$x = ym + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, C_N),$$

where ε is independent of y , together with ridge regularization $\rho(\theta) = \frac{\lambda}{2} \|\theta\|^2$. Taking $F = \text{span}(m)$, Theorem 7.1 gives

$$\mu_* = c_* (\lambda I_p + \nu_* C_N)^{-1} m \quad (18)$$

for some scalar c_* . Thus μ_* need not belong to $\text{span}(m)$ when the within-class covariance is anisotropic. The untransformed direction is recovered when $C_N m \in \text{span}(m)$, for instance when C_N is isotropic or m is an eigenvector of C_N .

Theorem 7.1 also isolates the source of possible non-Gaussianity in the test score. Indeed, by Theorem 6.1,

$$x^\top \hat{\theta} \stackrel{d}{\approx} x_I^\top \mu_* + \sigma_* z,$$

where $\sigma_*^2 := \mu_*^\top C_N \mu_* + \alpha_*^2$, and $z \sim \mathcal{N}(0, 1)$ is independent of (x_I, y) . Hence any non-Gaussian component of the asymptotic score must be carried by the structured projection $x_I^\top \mu_*$.

Software and Data

The code to reproduce our experiments is available at: <https://github.com/cosmital/Empirical-risk-minimization-asymptotics>.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Multiclass ERM asymptotics

Classification settings are simply characterized by the fact that $\mathcal{Y}$ is a finite set, then each possible value of $y \in \mathcal{Y}$ characterizes a different class. For simplicity, we assume here that $\mathcal{Y} = \{1, \dots, k\}$ where k is the number of classes and denote for all $\ell \in [k]$:

$$\begin{aligned}\gamma_\ell &:= \mathbb{P}(Y = \ell); & \mu_\ell &:= \mathbb{E}[x \mid y = \ell]; \\ C_\ell &:= \mathbb{E}[xx^\top \mid y = \ell] - \mu_\ell \mu_\ell^\top.\end{aligned}$$

Multiclass CGMT extensions are available for Gaussian designs (Mallory et al., 2025; Akhtiamov et al., 2024; Bosch & Panahi, 2025). We record the corresponding conditional extension of our equations.

Claim A.1 (Conditional multiclass formulation of ERM asymptotics). *Assume a multiclass analogue of Claim 4.1, local-uniform convergence to a unique stable saddle point, and Assumption 4. Then there exist unique $\mu_* \in \mathbb{R}^p$ and $\alpha, \kappa, \nu \in \mathbb{R}_+^k$ such that, for every $\ell \in [k]$,*

$$\begin{aligned}Q(\nu) &= \left(H_\rho + \sum_{h=1}^k \gamma_h \nu_h C_h \right)^{-1}, \\ \nu_\ell &= \frac{1}{\alpha_\ell \kappa_\ell} \mathbb{E}[z \xi_y(\mu_*^\top x + \alpha_\ell z, \kappa_\ell) \mid y = \ell], \\ \kappa_\ell &= \frac{1}{n} \text{tr}(C_\ell Q(\nu)), \\ \alpha_\ell^2 &= \sum_{h=1}^k \frac{\gamma_h \text{tr}(C_h Q(\nu) C_\ell Q(\nu))}{n \kappa_h^2} \mathbb{E}[\xi_y(\mu_*^\top x + \alpha_h z, \kappa_h)^2 \mid y = h], \\ 0 &= \nabla \rho(\mu_*) + \sum_{h=1}^k \frac{\gamma_h}{\kappa_h} \mathbb{E}[x \xi_y(\mu_*^\top x + \alpha_h z, \kappa_h) \mid y = h].\end{aligned}$$

Moreover, for every bounded Lipschitz $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[f(x^\top \hat{\theta}) \mid y = \ell] - \mathbb{E}[f(x^\top \mu_* + \alpha_\ell z) \mid y = \ell] \longrightarrow 0.$$

B. Proofs of Subsection 3 results

For simplicity, in the next proposition we keep the notation (X, Y) for deterministic objects. We define

$$F_{X,Y}(\theta) := \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{y_i}(x_i^\top \theta) + \rho(\theta), \quad \theta \in \mathbb{R}^p. \quad (19)$$

We will use multiple times the following classical stability lemma for minimizers.

Lemma B.1 (Stability of minimizers under strong convexity). *Let $\phi, \psi : \mathbb{R}^p \rightarrow \mathbb{R}$ be convex and $\mathcal{C}^1$. Assume moreover that ϕ is $\mathcal{C}^2$ and κ -strongly convex, i.e.*

$$\forall x \in \mathbb{R}^p, \quad \nabla^2 \phi(x) \succeq \kappa I_p \quad \text{for some } \kappa > 0.$$

Let $\mu_\phi := \arg \min_{x \in \mathbb{R}^p} \phi(x)$ (which is then unique). Then for any $x \in \mathbb{R}^p$,

$$\|\mu_\phi - x\|_2 \leq \frac{1}{\kappa} \|\nabla \phi(x)\|_2. \quad (20)$$

In particular, if $\mu_\psi := \arg \min_{x \in \mathbb{R}^p} \psi(x)$ exists and ψ is differentiable at μ_ψ , then

$$\|\mu_\phi - \mu_\psi\|_2 \leq \frac{1}{\kappa} \|(\nabla \phi - \nabla \psi)(\mu_\psi)\|_2 \leq \frac{1}{\kappa} \|\nabla \phi - \nabla \psi\|_{\mathcal{B}(0, \|\mu_\psi\|_2)}, \quad (21)$$

where, for any $f : \mathbb{R}^p \rightarrow \mathbb{R}^p$ and any $C > 0$,

$$\|f\|_{\mathcal{B}(0,C)} := \sup_{\|x\|_2 \leq C} \|f(x)\|_2.$$

In words, Lemma B.1 bounds the distance between minimizers via gradients, provided one of the objectives is uniformly strongly convex and one has a radius bound on the other minimizer.

Proof. Since $\nabla^2 \phi(x) \succeq \kappa I_p$ for all x , the gradient map $\nabla \phi$ is κ -strongly monotone, i.e., for all $x, x' \in \mathbb{R}^p$,

$$\langle \nabla \phi(x) - \nabla \phi(x'), x - x' \rangle \geq \kappa \|x - x'\|_2^2.$$

By Cauchy–Schwarz,

$$\kappa \|x - x'\|_2^2 \leq \|\nabla \phi(x) - \nabla \phi(x')\|_2 \|x - x'\|_2,$$

hence (if $x \neq x'$; otherwise the inequality is trivial)

$$\|x - x'\|_2 \leq \frac{1}{\kappa} \|\nabla \phi(x) - \nabla \phi(x')\|_2.$$

Taking $x' = \mu_\phi$ and using $\nabla \phi(\mu_\phi) = 0$ yields (20). If additionally μ_ψ is a minimizer of ψ and ψ is differentiable at μ_ψ (so $\nabla \psi(\mu_\psi) = 0$), then applying (20) with $x = \mu_\psi$ gives

$$\|\mu_\phi - \mu_\psi\|_2 \leq \frac{1}{\kappa} \|\nabla \phi(\mu_\psi)\|_2 = \frac{1}{\kappa} \|(\nabla \phi - \nabla \psi)(\mu_\psi)\|_2,$$

and the ball-sup bound follows immediately. $\square$

Proposition B.2 (Lipschitz dependence of $\hat{\theta}$ on (X, Y)). *For $(X, Y) \in \mathbb{R}^{p \times n} \times \mathcal{Y}^n$, define*

$$\hat{\theta}(X, Y) := \arg \min_{\theta \in \mathbb{R}^p} F_{X, Y}(\theta).$$

Under Assumption 2, assume that, for constants $B, C > 0$,

$$\|X\|_{\text{op}}, \|X'\|_{\text{op}} \leq C\sqrt{n}, \quad \frac{1}{n} \sum_{i=1}^n \left(|\mathcal{L}'_{y_i}(0)|^2 + |\mathcal{L}'_{y'_i}(0)|^2 \right) \leq B^2.$$

Then

$$\|\hat{\theta}(X, Y) - \hat{\theta}(X', Y')\|_2 \leq \frac{L}{\sqrt{n}} d((X, Y), (X', Y')),$$

for some constant L independent of n and p .

Proof. Since each $\mathcal{L}_{y_i}$ is convex and $\mathcal{C}^2$, we have $\mathcal{L}''_{y_i} \geq 0$. Since $\mathcal{L}'_{y_i}$ is λ -Lipschitz, we also have $\mathcal{L}''_{y_i} \leq \lambda$. Hence for all $\theta \in \mathbb{R}^p$,

$$\nabla^2 F_{X, Y}(\theta) = \frac{1}{n} X \text{diag}(\mathcal{L}''_{y_1}(x_1^\top \theta), \dots, \mathcal{L}''_{y_n}(x_n^\top \theta)) X^\top + \nabla^2 \rho(\theta) \succeq \kappa I_p, \quad (22)$$

so $F_{X, Y}$ is κ -strongly convex and admits a unique minimizer $\hat{\theta}(X, Y)$, characterized by $\nabla F_{X, Y}(\hat{\theta}(X, Y)) = 0$.

Step 1: a uniform bound on $\|\hat{\theta}(X, Y)\|_2$. Applying (20) (Lemma B.1) to $\phi = F_{X, Y}$ at the point $x = 0$ yields

$$\kappa \|\hat{\theta}(X, Y)\|_2 \leq \|\nabla F_{X, Y}(0)\|_2.$$

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $f(u)_i := \mathcal{L}'_{y_i}(u_i)$. By assumption, $n^{-1/2} \|f(0)\|_2 \leq B$. Set

$$B := \frac{1}{\sqrt{n}} \|f(0)\|_2, \quad R := \|\nabla \rho(0)\|_2.$$

Then

$$\|\nabla F_{X, Y}(0)\|_2 = \left\| \frac{1}{n} X f(0) + \nabla \rho(0) \right\|_2 \leq \frac{1}{n} \|X\|_{\text{op}} \|f(0)\|_2 + \|\nabla \rho(0)\|_2 \leq \frac{C\sqrt{n}}{n} \cdot \sqrt{n} B + R = CB + R.$$

Therefore,

$$\|\hat{\theta}(X, Y)\|_2 \leq \frac{1}{\kappa} (CB + R) =: M. \quad (23)$$

Step 2: stability of minimizers under perturbations of (X, Y) . Let $\theta := \hat{\theta}(X, Y)$ and $\theta' := \hat{\theta}(X', Y')$. Applying (21) (Lemma B.1) with $\phi = F_{X,Y}$ and $\psi = F_{X',Y'}$ gives

$$\|\theta - \theta'\|_2 \leq \frac{1}{\kappa} \|\nabla F_{X,Y} - \nabla F_{X',Y'}\|_{\mathcal{B}(0, \|\theta'\|_2)} \leq \frac{1}{\kappa} \|\nabla F_{X,Y} - \nabla F_{X',Y'}\|_{\mathcal{B}(0,M)}, \quad (24)$$

since $\|\theta'\|_2 \leq M$ by (23) applied to (X', Y') .

Fix an arbitrary $\bar{\theta} \in \mathbb{R}^p$ with $\|\bar{\theta}\|_2 \leq M$. We first bound the variation in X at fixed Y :

$$\nabla F_{X,Y}(\bar{\theta}) - \nabla F_{X',Y}(\bar{\theta}) = \frac{1}{n} \left(X f(X^\top \bar{\theta}) - X' f(X'^\top \bar{\theta}) \right) = \frac{1}{n} \left((X - X') f(X^\top \bar{\theta}) + X' (f(X^\top \bar{\theta}) - f(X'^\top \bar{\theta})) \right).$$

Hence, using $\|X'\|_{\text{op}} \leq C\sqrt{n}$,

$$\|\nabla F_{X,Y}(\bar{\theta}) - \nabla F_{X',Y}(\bar{\theta})\|_2 \leq \frac{1}{n} \|X - X'\|_{\text{op}} \|f(X^\top \bar{\theta})\|_2 + \frac{1}{n} \|X'\|_{\text{op}} \|f(X^\top \bar{\theta}) - f(X'^\top \bar{\theta})\|_2.$$

Since each coordinate $\mathcal{L}'_{y_i}$ is λ -Lipschitz, f is λ -Lipschitz on $(\mathbb{R}^n, \|\cdot\|_2)$, hence

$$\|f(X^\top \bar{\theta}) - f(X'^\top \bar{\theta})\|_2 \leq \lambda \|(X - X')^\top \bar{\theta}\|_2 \leq \lambda \|X - X'\|_{\text{op}} \|\bar{\theta}\|_2 \leq \lambda M \|X - X'\|_{\text{op}}.$$

Also,

$$\|f(X^\top \bar{\theta})\|_2 \leq \|f(0)\|_2 + \lambda \|X^\top \bar{\theta}\|_2 \leq \sqrt{n}B + \lambda \|X\|_{\text{op}} \|\bar{\theta}\|_2 \leq \sqrt{n}B + \lambda C\sqrt{n}M.$$

Combining,

$$\|\nabla F_{X,Y}(\bar{\theta}) - \nabla F_{X',Y}(\bar{\theta})\|_2 \leq \frac{1}{\sqrt{n}} \|X - X'\|_{\text{op}} (B + 2\lambda CM). \quad (25)$$

Second, we bound the variation in Y at fixed X' :

$$\|\nabla F_{X',Y}(\bar{\theta}) - \nabla F_{X',Y'}(\bar{\theta})\|_2 = \left\| \frac{1}{n} \sum_{i=1}^n \left(\mathcal{L}'_{y_i}(x'_i{}^\top \bar{\theta}) - \mathcal{L}'_{y'_i}(x'_i{}^\top \bar{\theta}) \right) x'_i \right\|_2 \quad (26)$$

$$\leq \frac{1}{n} \|X'\|_{\text{op}} \left(\sum_{i=1}^n \left(\mathcal{L}'_{y_i}(x'_i{}^\top \bar{\theta}) - \mathcal{L}'_{y'_i}(x'_i{}^\top \bar{\theta}) \right)^2 \right)^{1/2}. \quad (27)$$

Under the regularity assumption that $y \mapsto \mathcal{L}'_y(t)$ is λ -Lipschitz uniformly in t (with respect to d_Y),

$$\left| \mathcal{L}'_{y_i}(t) - \mathcal{L}'_{y'_i}(t) \right| \leq \lambda d_Y(y_i, y'_i).$$

Using $\|X'\|_{\text{op}} \leq C\sqrt{n}$ in (26) yields

$$\|\nabla F_{X',Y}(\bar{\theta}) - \nabla F_{X',Y'}(\bar{\theta})\|_2 \leq \frac{\lambda C}{\sqrt{n}} d_Y(Y, Y'), \quad d_Y(Y, Y') := \left(\sum_{i=1}^n d_Y(y_i, y'_i)^2 \right)^{1/2}.$$

Finally, combining the X and Y variations, inserting into (24), and using (23) gives

$$\|\hat{\theta}(X, Y) - \hat{\theta}(X', Y')\|_2 \leq \frac{1}{\kappa\sqrt{n}} \left((B + 2\lambda CM) \|X - X'\|_{\text{op}} + \lambda C d_Y(Y, Y') \right) \leq \frac{L}{\sqrt{n}} d((X, Y), (X', Y')),$$

for a constant L independent of n, p (absorbing κ, C, B, M, λ into L and using the definition of d). $\square$

Proof of Theorem 2.1. The samplewise Poincaré assumption implies the standard operator-norm bound

$$\| \|X\|_{\text{op}} \|_{L^q} = O_q(\sqrt{n}).$$

The moment condition in Assumption 2 also gives

$$\left\| \frac{1}{n} \sum_{i=1}^n |\mathcal{L}'_{y_i}(0)|^2 \right\|_{L^{q/2}} = O_q(1).$$

A truncation followed by a Lipschitz extension therefore reduces to Proposition B.2, so $(X, Y) \mapsto f(\hat{\theta}(X, Y))$ has block-gradient norm of order $n^{-1/2}$. Combining the tensorized L^q Poincaré inequality with this sensitivity estimate gives

$$\|f(\hat{\theta}) - \mathbb{E}f(\hat{\theta})\|_{L^q} \leq \frac{C_q}{\sqrt{n}},$$

which proves the claim. $\square$

Proof of Proposition 3.1 – quadratic approximation of ρ . Let $\mu_{\hat{\theta}} := \mathbb{E}[\hat{\theta}]$ and $\theta_0 := \hat{\theta} - \mu_{\hat{\theta}}$. A Taylor expansion of ρ at $\mu_{\hat{\theta}}$ with integral (Taylor–Lagrange) remainder yields

$$\left| \rho(\mu_{\hat{\theta}} + \theta_0) - \rho(\mu_{\hat{\theta}}) - \nabla \rho(\mu_{\hat{\theta}})^\top \theta_0 - \frac{1}{2} \theta_0^\top \nabla^2 \rho(\mu_{\hat{\theta}}) \theta_0 \right| \leq \frac{1}{6} \sup_{t \in [0, 1]} |\nabla^3 \rho(\mu_{\hat{\theta}} + t\theta_0) \cdot (\theta_0, \theta_0, \theta_0)|. \quad (28)$$

Using the decomposition of the third-order tensor into diagonal and off-diagonal parts, and the bound (valid for any $\theta, u \in \mathbb{R}^p$)

$$|\nabla^3 \rho(\theta) \cdot (u, u, u)| \leq \|\text{Diag}(\nabla^3 \rho(\theta))\|_F \|u^{(3)}\| + \|\text{OffDiag}(\nabla^3 \rho(\theta))\|_{\text{op}} \|u\|^3 \quad (29)$$

$$\leq C\sqrt{n} \|u^{(3)}\| + \frac{1}{\sqrt{n}} \|u\|^3, \quad (30)$$

(where we used $\|\text{Diag}(\nabla^3 \rho(\theta))\|_F \leq \sqrt{p} \|\nabla^3 \rho(\theta)\|_{\text{op}} \leq C\sqrt{n}$ under Assumption 3), we deduce from (28) that

$$\left| \rho(\hat{\theta}) - \rho(\mu_{\hat{\theta}}) - \nabla \rho(\mu_{\hat{\theta}})^\top \theta_0 - \frac{1}{2} \theta_0^\top \nabla^2 \rho(\mu_{\hat{\theta}}) \theta_0 \right| \leq C \max \left(\sqrt{n} \|\theta_0^{(3)}\|, \frac{1}{\sqrt{n}} \|\theta_0\|^3 \right). \quad (31)$$

By Theorem 2.1, for every coordinate j ,

$$\mathbb{E}|\theta_{0,j}|^6 = O(n^{-3}).$$

Since $p = O(n)$,

$$\mathbb{E}\|\theta_0^{(3)}\|^2 = \sum_{j=1}^p \mathbb{E}|\theta_{0,j}|^6 = O(n^{-2}).$$

Also $\mathbb{E}\|\theta_0\|^6 = O(1)$, using $\text{tr}(C_{\hat{\theta}}) = O(1)$ and the same L^q bound applied to $\theta \mapsto \|\theta - \mu_{\hat{\theta}}\|$. Taking expectations in (31) therefore gives

$$\mathbb{E}[R_n] = O(n^{-1/2}).$$

$\square$

Lemma B.3. Given a tensor $T \in \mathbb{R}^{p \times p \times p}$ and $u \in \mathbb{R}^p$, the contraction $T \cdot u \in \mathbb{R}^{p \times p}$ satisfies

$$\|T \cdot u\|_F \leq \sqrt{p} \|T \cdot u\|_{\text{op}} \leq \sqrt{p} \|T\|_{\text{op}} \|u\|.$$

For $\|\mu\| \leq M$, the resolvent identity and strong convexity give

$$\begin{aligned} \frac{1}{n} |\text{tr}(C_x(Q_\mu - Q_0))| &\leq \frac{1}{n} \|C_x\|_F \|Q_\mu\| \|\nabla^2 \rho(\mu) - H_\rho\|_F \|Q_0\| \\ &\leq C_M \frac{\sqrt{p}}{n} = O(n^{-1/2}). \end{aligned}$$

Proof of Lemma 3.2 – Constant approximation of $\nabla^2 \rho$. A Taylor–Lagrange argument for the Hessian gives

$$\|\nabla^2 \rho(\theta) - \nabla^2 \rho(0)\|_F \leq \sup_{t \in [0,1]} \|\nabla^3 \rho(t\theta) \cdot \theta\|_F. \quad (32)$$

Using Lemma B.3 and the decomposition into diagonal/off-diagonal terms,

$$\begin{aligned} \|\nabla^3 \rho(t\theta) \cdot \theta\|_F &\leq \|\text{Diag}(\nabla^3 \rho(t\theta)) \cdot \theta\|_F + \|\text{OffDiag}(\nabla^3 \rho(t\theta)) \cdot \theta\|_F \\ &\leq \|\text{Diag}(\nabla^3 \rho(t\theta))\|_{\text{op}} \|\theta\| + \sqrt{p} \|\text{OffDiag}(\nabla^3 \rho(t\theta))\|_{\text{op}} \|\theta\| \leq C \|\theta\|, \end{aligned}$$

by Assumption 3. Plugging into (32) yields

$$\|\nabla^2 \rho(\theta) - \nabla^2 \rho(0)\|_F \leq C \|\theta\|.$$

For $\|\mu\| \leq M$, the resolvent identity and strong convexity give

$$\begin{aligned} \frac{1}{n} |\text{tr}(C_x(Q_\mu - Q_0))| &\leq \frac{1}{n} \|C_x\|_F \|Q_\mu\| \|\nabla^2 \rho(\mu) - H_\rho\|_F \|Q_0\| \\ &\leq C_M \frac{\sqrt{p}}{n} = O(n^{-1/2}), \end{aligned}$$

using $p = O(n)$. $\square$

Proof of Theorem 3.3. At an interior saddle point, the fixed-point equations depend on the regularizer through H_ρ and $\nabla \rho(\mu_*)$. For the surrogate,

$$\nabla \rho_q(\mu_*) = \nabla \rho(\mu_{\hat{\theta}}) + H_\rho(\mu_* - \mu_{\hat{\theta}}) \longrightarrow \nabla \rho(\mu_*).$$

Uniqueness and stability therefore give the same limits (μ_*, α_*) for both ERMs. Since x is independent of the training data, set $\Sigma_x := \mathbb{E}[xx^\top]$. Then $\mathbb{E}[x^\top \hat{\theta}] = \mu_x^\top \mu_{\hat{\theta}}$ and

$$\mathbb{E}[(x^\top \hat{\theta})^2] = \mu_{\hat{\theta}}^\top \Sigma_x \mu_{\hat{\theta}} + \text{tr}(\Sigma_x C_{\hat{\theta}}).$$

Moreover, $\text{tr}(\Sigma_x C_{\hat{\theta}}) = \text{tr}(C_x C_{\hat{\theta}}) + \mu_x^\top C_{\hat{\theta}} \mu_x$, where the last term is $O(n^{-1})$. The claim follows from Theorem 4.3. $\square$

C. Conditional derivations for Subsection 4

Let us first introduce a result that explains where the resolvent comes from.

Lemma C.1 (Quadratic minimization on an ellipsoid: dual representation). *Let $p \geq 1$, let $H, C \in \mathbb{S}_{++}^p$ and $w \in \mathbb{R}^p$, and let $\alpha \geq 0$. Define the constrained quadratic program*

$$p(\alpha) := \min_{\theta \in \mathbb{R}^p} \left\{ w^\top \theta + \frac{1}{2} \theta^\top H \theta \quad : \quad \theta^\top C \theta = \alpha^2 \right\}. \quad (33)$$

Set the domain

$$\mathcal{D} := \{\nu \in \mathbb{R} : H + \nu C \succ 0\},$$

and for $\nu \in \mathcal{D}$ define

$$d(\nu) := -\frac{1}{2} w^\top (H + \nu C)^{-1} w - \frac{\alpha^2 \nu}{2}. \quad (34)$$

Then

$$p(\alpha) = \sup_{\nu \in \mathcal{D}} d(\nu). \quad (35)$$

Proof. The claim is trivial when $\alpha = 0$ since the constraint forces $\theta = 0$, hence $p(0) = 0$, and also $\sup_{\nu \in \mathcal{D}} d(\nu) = 0$ (take $\nu \rightarrow +\infty$ in (34)).

Assume $\alpha > 0$ in the remainder. Let $A := C^{-1/2}HC^{-1/2} \in \mathbb{S}_{++}^p$ and $\tilde{w} := C^{-1/2}w$. With the change of variables $\theta = \alpha C^{-1/2}z$, the constraint becomes $\|z\|^2 = 1$ and

$$w^\top \theta + \frac{1}{2} \theta^\top H \theta = \alpha \tilde{w}^\top z + \frac{\alpha^2}{2} z^\top A z.$$

Therefore,

$$p(\alpha) = \min_{\|z\|=1} \left\{ \alpha \tilde{w}^\top z + \frac{\alpha^2}{2} z^\top A z \right\}. \quad (36)$$

For $\lambda \in \mathbb{R}$, consider the Lagrangian

$$\mathcal{L}(z, \lambda) = \alpha \tilde{w}^\top z + \frac{\alpha^2}{2} z^\top A z + \frac{\lambda}{2} (\|z\|^2 - 1).$$

For any λ and any feasible z (i.e., $\|z\| = 1$) we have $\mathcal{L}(z, \lambda) = \alpha \tilde{w}^\top z + \frac{\alpha^2}{2} z^\top A z$, hence by taking the infimum over $z \in \mathbb{R}^p$,

$$\inf_{z \in \mathbb{R}^p} \mathcal{L}(z, \lambda) \leq p(\alpha). \quad (37)$$

If $\alpha^2 A + \lambda I \succ 0$, the function $z \mapsto \mathcal{L}(z, \lambda)$ is strictly convex and its unique minimizer satisfies

$$\nabla_z \mathcal{L}(z, \lambda) = \alpha \tilde{w} + (\alpha^2 A + \lambda I)z = 0 \implies z(\lambda) = -(\alpha^2 A + \lambda I)^{-1} \alpha \tilde{w}.$$

Substituting back yields

$$\inf_z \mathcal{L}(z, \lambda) = -\frac{\alpha^2}{2} \tilde{w}^\top (\alpha^2 A + \lambda I)^{-1} \tilde{w} - \frac{\lambda}{2}, \quad \text{for } \alpha^2 A + \lambda I \succ 0. \quad (38)$$

Since the feasible set $\{z : \|z\| = 1\}$ is nonempty and the objective in (36) is continuous, a minimizer z^* exists. The constraint is smooth and $\nabla(\|z\|^2 - 1) = 2z^* \neq 0$ on the sphere, hence the Lagrange multiplier theorem gives some $\lambda^* \in \mathbb{R}$ such that

$$\alpha \tilde{w} + \alpha^2 A z^* + \lambda^* z^* = 0.$$

If $\alpha^2 A + \lambda I$ has a negative eigenvalue, then $\inf_z \mathcal{L}(z, \lambda) = -\infty$, so such λ do not contribute to the dual supremum. Therefore it suffices to consider λ such that $\alpha^2 A + \lambda I \succeq 0$, and we first treat the case $\succ 0$.

If $\alpha^2 A + \lambda^* I \succ 0$, then $z^* = z(\lambda^*)$ and therefore

$$p(\alpha) = \mathcal{L}(z^*, \lambda^*) = \inf_z \mathcal{L}(z, \lambda^*).$$

Combining with (37)–(38) yields

$$p(\alpha) = \sup_{\lambda: \alpha^2 A + \lambda I \succ 0} \left\{ -\frac{\alpha^2}{2} \tilde{w}^\top (\alpha^2 A + \lambda I)^{-1} \tilde{w} - \frac{\lambda}{2} \right\}. \quad (39)$$

If instead $\alpha^2 A + \lambda^* I \succeq 0$ is singular, the same identity holds with a supremum by a standard limiting argument (approximate λ^* from above so that $\alpha^2 A + \lambda I \succ 0$ and pass to the limit); this is why we state (35) with “sup”.

Let $\nu := \lambda/\alpha^2$. Then $\alpha^2 A + \lambda I = \alpha^2(A + \nu I)$ and, using

$$H + \nu C = C^{1/2}(A + \nu I)C^{1/2}, \quad (H + \nu C)^{-1} = C^{-1/2}(A + \nu I)^{-1}C^{-1/2},$$

we obtain

$$\tilde{w}^\top (A + \nu I)^{-1} \tilde{w} = w^\top (H + \nu C)^{-1} w.$$

Thus (39) becomes exactly (35) with $d(\nu)$ as in (34). □

Proposition C.2 (Formal resolution of the auxiliary problem). *Let $\mu, x \in \mathbb{R}^p$, $h \in \mathbb{R}^n$, and let $C, H \in \mathbb{R}^{p \times p}$ be symmetric positive definite. Let $\mathcal{L} : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ be proper, lower semicontinuous, and convex. Assume that the saddle-point problem below admits a unique saddle point $(\tilde{\theta}, \tilde{w})$ and that the max–min interchanges used in the derivation are valid:*

$$(\tilde{\theta}, \tilde{w}) := \arg \min_{\theta \in \mathbb{R}^p} \max_{w \in \mathbb{R}^n} \left\{ \|C^{1/2}\theta\| w^\top h + (\mu^\top x) w^\top \mathbf{1} + \|w\| \theta^\top x - \frac{1}{n} \mathcal{L}^*(nw) + \frac{1}{2} \theta^\top H \theta \right\}. \quad (40)$$

Denote $\tilde{\alpha} := \|C^{1/2}\tilde{\theta}\|$ and $\tilde{\beta} := \sqrt{n}\|\tilde{w}\|$. Then $(\tilde{\alpha}, \tilde{\kappa}, \tilde{\beta}, \tilde{\nu})$ can be characterized through the saddle point

$$(\tilde{\alpha}, \tilde{\kappa}, \tilde{\beta}, \tilde{\nu}) = \arg \min_{\alpha \geq 0, \kappa > 0} \arg \max_{\beta \geq 0, \nu \in \mathcal{D}} \left\{ \frac{\beta^2 \kappa}{2} + \frac{1}{n} e_{\mathcal{L}}(\alpha h + (\mu^\top x) \mathbf{1}; \kappa) - \frac{\beta^2}{2n} x^\top (H + \nu C)^{-1} x - \frac{\alpha^2 \nu}{2} \right\}, \quad (41)$$

where $\mathcal{D} := \{\nu \in \mathbb{R} : H + \nu C \succ 0\}$.

Proof. Since $\mathcal{L}$ is proper l.s.c. convex, the Fenchel–Moreau theorem yields

$$\mathcal{L}^*(z) = \sup_{v \in \mathbb{R}^n} \{v^\top z - \mathcal{L}(v)\} \implies -\frac{1}{n} \mathcal{L}^*(nw) = \min_{v \in \mathbb{R}^n} \left\{ -w^\top v + \frac{1}{n} \mathcal{L}(v) \right\}.$$

We thus have the identity:

$$(\tilde{\theta}, \tilde{w}, \tilde{\nu}) \in \arg \min_{\theta} \max_w \min_v \left\{ \|C^{1/2}\theta\| w^\top h + (\mu^\top x) w^\top \mathbf{1} - w^\top v + \|w\| \theta^\top x + \frac{1}{n} \mathcal{L}(v) + \frac{1}{2} \theta^\top H \theta \right\}. \quad (42)$$

Write $w = (\beta/\sqrt{n})u$ with $\beta := \sqrt{n}\|w\| \geq 0$ and $u \in \mathbb{S}^{n-1}$. For fixed (θ, β) , consider the inner game in (u, v) :

$$\phi_s(u, v) := \frac{\beta}{\sqrt{n}} u^\top (\|C^{1/2}\theta\| h + (\mu^\top x) \mathbf{1} - v) + \frac{1}{n} \mathcal{L}(v).$$

Under the max–min interchange and convexification assumed in the proposition, the directional optimization yields

$$\max_{\|u\| \leq 1} \frac{\beta}{\sqrt{n}} u^\top (\|C^{1/2}\theta\| h + (\mu^\top x) \mathbf{1} - v) = \frac{\beta}{\sqrt{n}} \|\|C^{1/2}\theta\| h + (\mu^\top x) \mathbf{1} - v\|.$$

Consequently,

$$\max_u \min_v \phi_s(u, v) = \min_{v \in \mathbb{R}^n} \left\{ \frac{\beta}{\sqrt{n}} \|\|C^{1/2}\theta\| h + (\mu^\top x) \mathbf{1} - v\| + \frac{1}{n} \mathcal{L}(v) \right\}.$$

Plugging this into (42) yields

$$(\tilde{\theta}, \tilde{\beta}) = \arg \min_{\theta} \max_{\beta \geq 0} \left\{ \phi_{\mathcal{L}}(\|C^{1/2}\theta\|, \beta) + \frac{\beta}{\sqrt{n}} \theta^\top x + \frac{1}{2} \theta^\top H \theta \right\}, \quad (43)$$

where, for $\alpha \geq 0$ and $\beta \geq 0$,

$$\phi_{\mathcal{L}}(\alpha, \beta) := \min_{v \in \mathbb{R}^n} \left\{ \frac{\beta}{\sqrt{n}} \|\alpha h + (\mu^\top x) \mathbf{1} - v\| + \frac{1}{n} \mathcal{L}(v) \right\}.$$

Let $u := \alpha h + (\mu^\top x) \mathbf{1}$. Using the standard identity $t = \min_{\tau > 0} \{\tau/2 + t^2/(2\tau)\}$ for $t \geq 0$, we obtain

$$\begin{aligned} \phi_{\mathcal{L}}(\alpha, \beta) &= \min_v \left\{ \frac{\beta}{\sqrt{n}} \|u - v\| + \frac{1}{n} \mathcal{L}(v) \right\} \\ &= \min_{\tau_{\mathcal{L}} > 0, v} \left\{ \frac{\beta \tau_{\mathcal{L}}}{2} + \frac{\beta}{2\tau_{\mathcal{L}} n} \|u - v\|^2 + \frac{1}{n} \mathcal{L}(v) \right\}. \end{aligned}$$

Set $\kappa := \tau_{\mathcal{L}}/\beta$ (with the convention that κ is optimized only when $\beta > 0$). Then $\beta\tau_{\mathcal{L}}/2 = \beta^2\kappa/2$ and $\beta/(2\tau_{\mathcal{L}}) = 1/(2\kappa)$, hence

$$\phi_{\mathcal{L}}(\alpha, \beta) = \min_{\kappa > 0} \left\{ \frac{\beta^2\kappa}{2} + \frac{1}{n}e_{\mathcal{L}}(u; \kappa) \right\}, \quad e_{\mathcal{L}}(u; \kappa) := \min_{v \in \mathbb{R}^n} \left\{ \frac{1}{2\kappa}\|u - v\|^2 + \mathcal{L}(v) \right\}. \quad (44)$$

Injecting (44) into (43) gives

$$(\tilde{\theta}, \tilde{\beta}) = \arg \min_{\theta} \max_{\beta \geq 0} \min_{\kappa > 0} \left\{ \frac{\beta^2\kappa}{2} + \frac{1}{n}e_{\mathcal{L}} \left(\|C^{1/2}\theta\| h + (\mu^\top x)\mathbf{1}; \kappa \right) + \frac{\beta}{\sqrt{n}}\theta^\top x + \frac{1}{2}\theta^\top H\theta \right\}. \quad (45)$$

Let $\alpha := \|C^{1/2}\theta\| \geq 0$. For fixed (α, β, κ) , the dependence in θ inside (45) is through

$$\frac{\beta}{\sqrt{n}}x^\top \theta + \frac{1}{2}\theta^\top H\theta \quad \text{under the constraint} \quad \theta^\top C\theta = \alpha^2.$$

Lemma C.1 yields

$$\min_{\theta: \theta^\top C\theta = \alpha^2} \left\{ \frac{\beta}{\sqrt{n}}x^\top \theta + \frac{1}{2}\theta^\top H\theta \right\} = \max_{\nu \in \mathcal{D}} \left\{ -\frac{\beta^2}{2n}x^\top (H + \nu C)^{-1}x - \frac{\alpha^2\nu}{2} \right\}.$$

Combining this with (45) and using that the remaining term depends on θ only through α , we obtain the claimed characterization (41). $\square$

Lemma C.3. *Let E be a finite-dimensional real vector space, let Z be a random variable with values in $\mathcal{Z}$, and for each $z \in \mathcal{Z}$ let $\psi_z : E \rightarrow \mathbb{R}$ be C^1 , strictly convex, and admitting a unique minimizer. Define the random minimizer*

$$\hat{u} := \arg \min_{u \in E} \psi_Z(u).$$

Assume that $\hat{u}$ is integrable and that for all $m \in E$ the quantity $\mathbb{E}[\psi_Z(\hat{u} - \mathbb{E}[\hat{u}] + m)]$ is finite. Then $\mathbb{E}[\hat{u}]$ is the (unique) minimizer of

$$m \mapsto \mathbb{E}[\psi_Z((\hat{u} - \mathbb{E}[\hat{u}]) + m)],$$

i.e.

$$\mathbb{E}[\hat{u}] = \arg \min_{m \in E} \mathbb{E}[\psi_Z((\hat{u} - \mathbb{E}[\hat{u}]) + m)].$$

Proof. For each realization $Z = z$, the point $\hat{u} = \hat{u}(z)$ minimizes ψ_z , hence $\nabla \psi_z(\hat{u}) = 0$. By convexity, for any $m \in E$,

$$\psi_z(\hat{u} - \mathbb{E}[\hat{u}] + m) \geq \psi_z(\hat{u}) + \langle \nabla \psi_z(\hat{u}), m - \mathbb{E}[\hat{u}] \rangle = \psi_z(\hat{u}).$$

Taking expectations yields

$$\mathbb{E}[\psi_Z(\hat{u} - \mathbb{E}[\hat{u}] + m)] \geq \mathbb{E}[\psi_Z(\hat{u})] = \mathbb{E}[\psi_Z(\hat{u} - \mathbb{E}[\hat{u}] + \mathbb{E}[\hat{u}])],$$

so $m = \mathbb{E}[\hat{u}]$ is a minimizer. Uniqueness follows from strict convexity. $\square$

Corollary C.4. *Under Assumption 2, letting $\hat{\theta}_0 = \hat{\theta} - \mu_{\hat{\theta}}$, we have*

$$\mu_{\hat{\theta}} = \arg \min_{\mu \in \mathbb{R}^p} \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \mathcal{L}_{y_i}(x_i^\top (\hat{\theta}_0 + \mu)) + \rho(\hat{\theta}_0 + \mu) \right].$$

Proposition C.5 (Conditional concentration of the auxiliary radius). *Consider the auxiliary saddle problem in (41), restricted to a fixed compact saddle domain. Suppose its profiled objective is uniformly τ -strongly convex in α , its α -gradient has $O(n^{-1/2})$ sensitivity to the auxiliary randomness, and its deterministic profiles converge locally uniformly to a limit with a unique stable α -minimizer α_* . For every fixed $q \geq 2$,*

$$\|\tilde{\alpha} - \mathbb{E}\tilde{\alpha}\|_{L^q} \leq \frac{C_q}{\sqrt{n}}, \quad \mathbb{E}\tilde{\alpha} \longrightarrow \alpha_*. \quad (46)$$

In particular, $\tilde{\alpha} \xrightarrow{\mathbb{P}} \alpha_*$. Here α_* is the α -component of the deterministic saddle problem

$$\min_{\alpha \geq 0, \kappa > 0} \max_{\beta \geq 0, \nu \in \mathcal{D}} \left\{ \frac{\beta^2 \kappa}{2} + \frac{1}{n} \mathbb{E} [e_{\mathcal{L}}(\alpha g + (\mu^\top x) \mathbf{1}; \kappa)] - \frac{\beta^2}{2n} \text{tr}(C_x (H + \nu C)^{-1}) - \frac{\alpha^2 \nu}{2} \right\}.$$

Proof. Let j_n denote the random objective after profiling the other saddle variables. Strong convexity and Lemma B.1 turn the assumed gradient-sensitivity bound into

$$|\tilde{\alpha}(Z) - \tilde{\alpha}(Z')| \leq \frac{L}{\tau \sqrt{n}} d(Z, Z'),$$

where Z collects the auxiliary randomness. The tensorized L^q Poincaré inequality therefore gives

$$\|\tilde{\alpha} - \mathbb{E}\tilde{\alpha}\|_{L^q} \leq C_q n^{-1/2}.$$

Local-uniform convergence of the profiles and stability of the unique limiting minimizer imply $\tilde{\alpha} \rightarrow \alpha_*$ in probability. Compactness gives uniform integrability, hence $\mathbb{E}\tilde{\alpha} \rightarrow \alpha_*$ as well. $\square$

Conditional derivation of Theorem 4.3. Apply Claim 4.1 to the centered decomposition (11), after the compact truncation stated in the main text. Its auxiliary optimization is exactly the problem of Proposition C.2, with the quadratic replacement justified by Proposition 3.1 and Lemma 3.2.

By the uniform-integrability and local-uniform concentration hypotheses of the theorem, the random auxiliary saddle objectives converge to $\mathcal{J}$. Stability of its unique saddle point therefore gives

$$\|C_x^{1/2}(\hat{\theta} - \mu_{\hat{\theta}})\| \xrightarrow{\mathbb{P}} \alpha_*.$$

The concentration from Theorem 2.1 gives uniform integrability of its square, hence $\text{tr}(C_x C_{\hat{\theta}}) \rightarrow \alpha_*^2$.

For the mean, Corollary C.4 represents $\mu_{\hat{\theta}}$ as the minimizer of the expected shifted primal objective. The comparison principle, uniform integrability, and local-uniform convergence identify its limit with the μ -profile of $\mathcal{J}$. Strong convexity and saddle-point stability then imply $\|\mu_{\hat{\theta}} - \mu_*\| \rightarrow 0$. $\square$

Proof of Theorem 4.4 – Fixed-point formulation. Basic properties of the Moreau envelope yield

$$\partial_u e_{\mathcal{L}_y}(u; \kappa) = \frac{1}{\kappa} \xi_y(u, \kappa), \quad \partial_\kappa e_{\mathcal{L}_y}(u, \kappa) = -\frac{1}{2\kappa^2} \xi_y(u, \kappa)^2.$$

Treat μ, α, κ as minimization variables and β, ν as maximization variables in (4) and denote $u := x^\top \mu + \alpha z$. The stationarity conditions are:

- Gradient with respect to μ :

$$\nabla \rho(\mu) + \frac{1}{\kappa} \mathbb{E}[\xi_y(u, \kappa) x] = 0.$$

- Gradient with respect to α :

$$\frac{1}{\kappa} \mathbb{E}[\xi_y(u, \kappa) z] - \nu \alpha = 0 \implies \nu \alpha = \frac{1}{\kappa} \mathbb{E}[\xi_y(u, \kappa) z].$$

- Gradient with respect to κ :

$$\frac{\beta^2}{2} - \frac{1}{2\kappa^2} \mathbb{E}[\xi_y(u, \kappa)^2] = 0 \implies \beta^2 = \frac{\mathbb{E}[\xi_y(u, \kappa)^2]}{\kappa^2}.$$

- Gradient with respect to β :

$$\beta \left(\kappa - \frac{1}{n} \text{tr}(C_x Q(\nu)) \right) = 0 \implies \kappa = \frac{1}{n} \text{tr}(C_x Q(\nu)),$$

since $\beta > 0$ at the optimum.

- Gradient with respect to ν . Using $\frac{d}{d\nu}Q(\nu) = -Q(\nu)C_xQ(\nu)$, we have

$$\frac{d}{d\nu}\text{tr}(C_xQ(\nu)) = -\text{tr}(C_xQ(\nu)C_xQ(\nu)) = -nA(\nu),$$

with

$$A(\nu) := \frac{1}{n}\text{tr}(C_xQ(\nu)C_xQ(\nu)).$$

Then stationarity gives

$$-\frac{\alpha^2}{2} - \frac{\beta^2}{2n} \frac{d}{d\nu}\text{tr}(C_xQ(\nu)) = 0 \implies \frac{\alpha^2}{\beta^2} = A(\nu).$$

Combining the last three identities to eliminate α and β yields

$$\nu^2 A(\nu) \mathbb{E}[\xi_y(u, \kappa)^2] = (\mathbb{E}[\xi_y(u, \kappa)z])^2.$$

□

D. Proofs of Section 5 results and Link with random matrix theory methods

D.1. Proof of Corollary 5.1

For $\mathcal{L}_y(t) = \frac{1}{2}(t - y)^2$,

$$\text{prox}_{\kappa\mathcal{L}_y}(u) = \frac{u + \kappa y}{1 + \kappa}, \quad \xi_y(u, \kappa) = \frac{\kappa}{1 + \kappa}(u - y).$$

Let $u = \mu^\top x + \alpha z$ and $\gamma = \kappa/(1 + \kappa)$. Since ε is centered and independent of (x, z) ,

$$\begin{aligned} \mathbb{E}[\xi_y(u, \kappa)z] &= \gamma\alpha, \\ \mathbb{E}[\xi_y(u, \kappa)x] &= \gamma\Sigma_x(\mu - \theta^*), \\ \mathbb{E}[\xi_y(u, \kappa)^2] &= \gamma^2((\mu - \theta^*)^\top \Sigma_x(\mu - \theta^*) + \alpha^2 + \sigma_\varepsilon^2). \end{aligned} \tag{47}$$

The second fixed-point equation gives

$$\nu_* = \frac{\gamma_*}{\kappa_*} = \frac{1}{1 + \kappa_*} = \frac{1}{1 + \frac{1}{n}\text{tr}(C_xQ(\nu_*))},$$

whereas the vector equation gives

$$\mu_* = (H + \nu_*\Sigma_x)^{-1}(\nu_*\Sigma_x\theta^* - a) = \mu(\nu_*).$$

Moreover, independence of the test sample and $\hat{\theta}$ yields

$$\begin{aligned} \mathbb{E}[\mathcal{L}_y(x^\top \hat{\theta})] &= \frac{1}{2}((\mu_{\hat{\theta}} - \theta^*)^\top \Sigma_x(\mu_{\hat{\theta}} - \theta^*) + \text{tr}(\Sigma_x C_{\hat{\theta}}) + \sigma_\varepsilon^2) \\ &= \frac{1}{2}((\mu_{\hat{\theta}} - \theta^*)^\top \Sigma_x(\mu_{\hat{\theta}} - \theta^*) + \text{tr}(C_x C_{\hat{\theta}}) + \mu_x^\top C_{\hat{\theta}} \mu_x + \sigma_\varepsilon^2) \\ &\longrightarrow \frac{1}{2}(\Delta(\nu_*) + \alpha_*^2 + \sigma_\varepsilon^2). \end{aligned} \tag{48}$$

Here, the first quadratic term equals

$$(\mu_{\hat{\theta}} - \theta^*)^\top C_x(\mu_{\hat{\theta}} - \theta^*) + (\mu_x^\top (\mu_{\hat{\theta}} - \theta^*))^2,$$

so the nonzero-mean bias is retained, whereas

$$\mu_x^\top C_{\hat{\theta}} \mu_x \leq \|\mu_x\|^2 \|C_{\hat{\theta}}\| = O(n^{-1}).$$

Finally, (47) and the third fixed-point equation give

$$\alpha_*^2 = \frac{\nu_*^2 A(\nu_*) (\Delta(\nu_*) + \sigma_\varepsilon^2)}{1 - \nu_*^2 A(\nu_*)}.$$

Consequently,

$$\Delta(\nu_*) + \alpha_*^2 + \sigma_\varepsilon^2 = \frac{\Delta(\nu_*) + \sigma_\varepsilon^2}{1 - \nu_*^2 A(\nu_*)},$$

which proves the corollary.

D.2. RMT consistency check for centered ridge regression

We retain here a direct random-matrix derivation to show how the parameters in Corollary 5.1 arise from classical resolvent deterministic equivalents. To keep the comparison transparent, this subsection specializes to $\mathbb{E}[x] = 0$, $a = 0$, and $H = \lambda I_p$, so that $\rho(\theta) = \frac{\lambda}{2} \|\theta\|^2$. With $X = (x_1, \dots, x_n) \in \mathbb{R}^{p \times n}$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^\top$, the ridge problem is

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^p} \frac{1}{2n} \|X^\top(\theta - \theta^*) - \varepsilon\|^2 + \frac{\lambda}{2} \|\theta\|^2. \quad (49)$$

Its minimizer has the closed form

$$\hat{\theta} = R \left(\frac{1}{n} X X^\top \theta^* + \frac{1}{n} X \varepsilon \right) = (I_p - \lambda R) \theta^* + \frac{1}{n} R X \varepsilon, \quad (50)$$

where

$$R := \left(\frac{1}{n} X X^\top + \lambda I_p \right)^{-1}.$$

Recall $Q(\nu) := (\nu C_x + \lambda I_p)^{-1}$, and let $\nu_* > 0$ be the unique solution of

$$\frac{1}{\nu} = 1 + \frac{1}{n} \text{tr}(C_x Q(\nu)),$$

which is the same equation as (15). Standard leave-one-out resolvent arguments give $Q(\nu_*)$ as a deterministic equivalent of R . In what follows, $M \simeq \bar{M}$ means that $\text{tr}(S(M - \bar{M})) \xrightarrow{\mathbb{P}} 0$ for every deterministic test matrix S with uniformly bounded Frobenius norm. In this sense,

$$R \simeq Q(\nu_*). \quad (51)$$

For a deterministic matrix $B \in \mathbb{R}^{p \times p}$ with bounded operator norm, the corresponding second-order deterministic equivalent is

$$Q_2(B) \equiv Q B Q + \frac{\nu^2 \text{tr}(C_x Q B Q)}{1 - \nu^2 A(\nu)} Q C_x Q, \quad (52)$$

where $Q = Q(\nu)$ and, as in Corollary 5.1,

$$A(\nu) := \frac{1}{n} \text{tr}(Q C_x Q C_x)$$

one has $R B R \simeq Q_2(B)$. We henceforth evaluate all deterministic quantities at $\nu = \nu_*$.

D.3. Mean and covariance from the RMT equivalents

Since ε is independent of X and centered, (50) gives

$$\mu_{\hat{\theta}} = \mathbb{E}[(I_p - \lambda R) \theta^*] \simeq (I_p - \lambda Q) \theta^*, \quad (53)$$

where $RXX^\top/n = I_p - \lambda R$. Thus the RMT mean coincides with the specialization of $\mu(\nu_*)$ in Corollary 5.1.

To estimate the covariance, set $M := I_p - \lambda R$. Independence gives

$$\mathbb{E}[\hat{\theta}\hat{\theta}^\top] = \mathbb{E}[M\theta^*\theta^{*\top}M] + \frac{\sigma_\varepsilon^2}{n^2}\mathbb{E}[RXX^\top R].$$

We first evaluate the noise contribution. For a deterministic test matrix $S \in \mathbb{R}^{p \times p}$,

$$\begin{aligned} \frac{1}{n^2}\mathbb{E}[\varepsilon^\top X^\top RSRX\varepsilon] &= \frac{\sigma_\varepsilon^2}{n^2}\mathbb{E}\text{tr}(RXX^\top RS) \\ &= \frac{\sigma_\varepsilon^2}{n}\mathbb{E}\text{tr}((R - \lambda R^2)S) \simeq \frac{\sigma_\varepsilon^2\nu^2}{n} \frac{\text{tr}(QC_xQS)}{1 - \nu^2 A(\nu)}. \end{aligned}$$

The last equivalence follows from (52) and the identities

$$\lambda Q = I_p - \nu C_x Q, \quad \frac{\lambda}{n}\text{tr}(QC_x Q) = \frac{1}{n}\text{tr}(C_x Q) - \nu A(\nu), \quad 1 - \frac{\nu}{n}\text{tr}(C_x Q) = \nu.$$

For the design-induced contribution, expanding $MSM = S - \lambda(RS + SR) + \lambda^2 RSR$ and applying (52) shows that

$$\mathbb{E}[MSM] - \mathbb{E}[M]S\mathbb{E}[M] \simeq \frac{\nu^2\lambda^2}{n} \frac{\text{tr}(C_x QSQ)}{1 - \nu^2 A(\nu)} QC_x Q.$$

Taking $S = \theta^*\theta^{*\top}$ and combining the two contributions therefore gives the covariance deterministic equivalent

$$C_{\hat{\theta}} \simeq \frac{\nu^2}{n} \frac{\sigma_\varepsilon^2 + \lambda^2 \theta^{*\top} QC_x Q \theta^*}{1 - \nu^2 A(\nu)} QC_x Q.$$

Finally, (53) gives $\mu_{\hat{\theta}} - \theta^* \simeq -\lambda Q\theta^*$. Hence

$$\text{tr}(C_x C_{\hat{\theta}}) \simeq \frac{\nu^2 A(\nu)}{1 - \nu^2 A(\nu)} [\sigma_\varepsilon^2 + (\mu_{\hat{\theta}} - \theta^*)^\top C_x (\mu_{\hat{\theta}} - \theta^*)].$$

This is precisely the formula for α_*^2 obtained from the min-max fixed-point equations in the centered isotropic-ridge setting. The calculation therefore retains the original RMT route while making explicit its correspondence with Corollary 5.1.

E. Proofs of Subsection 6 results

Proof of Theorem 6.1. Let

$$g \sim \mathcal{N}(\mu_{\hat{\theta}}, C_{\hat{\theta}})$$

be independent of (x, y) . By Assumption 4 and $\mathbb{E}\|x\|/\sqrt{n} = O(1)$,

$$W_1(x^\top(\hat{\theta} - \mu_{\hat{\theta}}), x^\top(g - \mu_{\hat{\theta}})) = o(1).$$

Conditionally on x ,

$$x^\top g = x^\top \mu_{\hat{\theta}} + \sqrt{x^\top C_{\hat{\theta}} x} z.$$

Applying the Poincaré inequality to $q_A(x) = x^\top Ax$ gives

$$\text{Var}(q_A(x)) \leq C_P \mathbb{E}\|\nabla q_A(x)\|^2 = 4C_P \mathbb{E}\|Ax\|^2 \leq C\|A\|_F^2.$$

For $A = C_{\hat{\theta}}$,

$$\|A\|_F^2 \leq p\|A\|^2 = O(n^{-1})$$

by Corollary 2.2. Hence

$$x^\top C_{\hat{\theta}} x - \text{tr}(\Sigma_x C_{\hat{\theta}}) \xrightarrow{\mathbb{P}} 0.$$

Furthermore,

$$\text{tr}(\Sigma_x C_{\hat{\theta}}) = \text{tr}(C_x C_{\hat{\theta}}) + \mu_x^\top C_{\hat{\theta}} \mu_x = \alpha_*^2 + o(1),$$

and

$$\mathbb{E} \left[(x^\top (\mu_{\hat{\theta}} - \mu_*))^2 \right] \leq \|\Sigma_x\| \|\mu_{\hat{\theta}} - \mu_*\|^2 \longrightarrow 0.$$

Theorem 4.3 and the uniform Lipschitz property of f complete the proof. $\square$

F. Proofs of Section 7 results.

Proof of Theorem 7.1. Set

$$U_* := \mu_*^\top x + \alpha_* z, \quad \Xi_* := \xi_y(U_*, \kappa_*).$$

The vector equation in Theorem 4.4 reads

$$0 = \nabla \rho(\mu_*) + \frac{1}{\kappa_*} \mathbb{E}[x_I \Xi_*] + \frac{1}{\kappa_*} \mathbb{E}[x_N \Xi_*]. \quad (54)$$

Since $x_I \in F$ almost surely, one has

$$\mathbb{E}[x_I \Xi_*] \in F. \quad (55)$$

It remains to identify the contribution of x_N . The map $u \mapsto \xi_y(u, \kappa_*)$ is Lipschitz; under Assumption 2, it is continuously differentiable with bounded derivative. Since $x_N \sim \mathcal{N}(0, C_N)$ is independent of (x_I, y, z) , Gaussian integration by parts, conditionally on (x_I, y, z) , gives

$$\mathbb{E}[x_N \Xi_*] = C_N \mu_* \mathbb{E}[\partial_u \xi_y(U_*, \kappa_*)]. \quad (56)$$

Applying the one-dimensional Gaussian integration-by-parts identity to z similarly yields

$$\mathbb{E}[z \Xi_*] = \alpha_* \mathbb{E}[\partial_u \xi_y(U_*, \kappa_*)]. \quad (57)$$

The scalar fixed-point equation for ν_* therefore implies

$$\nu_* = \frac{1}{\alpha_* \kappa_*} \mathbb{E}[z \Xi_*] = \frac{1}{\kappa_*} \mathbb{E}[\partial_u \xi_y(U_*, \kappa_*)]. \quad (58)$$

Combining (56)–(58), we obtain

$$\frac{1}{\kappa_*} \mathbb{E}[x_N \Xi_*] = \nu_* C_N \mu_*. \quad (59)$$

Substituting (59) into (54) gives

$$\nabla \rho(\mu_*) + \nu_* C_N \mu_* = -\frac{1}{\kappa_*} \mathbb{E}[x_I \Xi_*] \in F, \quad (60)$$

which proves the first assertion of the theorem.

Now let $H_\rho = \nabla^2 \rho(0)$ and $a_* = \nabla \rho(\mu_*) - H_\rho \mu_*$. Rewriting (60) gives

$$(H_\rho + \nu_* C_N) \mu_* = -a_* - \frac{1}{\kappa_*} \mathbb{E}[x_I \Xi_*] \in F + \text{span}(a_*). \quad (61)$$

Assumption 2 gives $H_\rho \succ 0$, while $C_N \succeq 0$; hence $H_\rho + \nu_* C_N$ is invertible. Therefore

$$\mu_* \in (H_\rho + \nu_* C_N)^{-1} (F + \text{span}(a_*)) = \mathcal{S}_*.$$

For a quadratic regularizer, $H_\rho = H$ and $a_* = a$, so the confinement space becomes $(H + \nu_* C_N)^{-1} (F + \text{span}(a))$. Finally, Theorem 4.3 gives $\|\mu_{\hat{\theta}} - \mu_*\| \rightarrow 0$, and therefore

$$\text{dist}(\mu_{\hat{\theta}}, \mathcal{S}_*) \leq \|\mu_{\hat{\theta}} - \mu_*\| \xrightarrow{n \rightarrow \infty} 0.$$

$\square$

F.1. Recovery of the Mai–Liao LFMM equations

We finally make explicit how the fixed-point system above recovers the linear factor mixture model (LFMM) equations of Mai & Liao (2025). Their binary model is

$$x = \sum_{k=1}^p (ys_k + e_k)v_k, \quad s_k = 0 \quad \text{for } k > q, \quad (62)$$

where $y \in \{-1, 1\}$, the v_k are linearly independent, and the e_k are independent, centered, unit-variance factors. The number q of informative factors is fixed. In their notation,

$$\mu := \mathbb{E}[x \mid y = 1] = \sum_{k=1}^q s_k v_k, \quad \Sigma := \text{Cov}(x \mid y) = \sum_{k=1}^p v_k v_k^\top,$$

and

$$h_\kappa(t, y) := \frac{1}{\kappa} \left(\text{prox}_{\kappa \mathcal{L}_y}(t) - t \right). \quad (63)$$

For reference, their scalar system can be written as

$$\begin{aligned} Q &= (\lambda I_p + \vartheta \Sigma)^{-1}, & \kappa &= \frac{1}{n} \text{tr}(\Sigma Q), & \vartheta &= -\mathbb{E}[\partial_r h_\kappa(r, y)], \\ \gamma^2 &= \mathbb{E}[h_\kappa(r, y)^2], & \eta &= \mathbb{E}[y h_\kappa(r, y)], & \omega_k &= \mathbb{E}[e_k h_\kappa(r, y)] + \vartheta v_k^\top Q \xi, \\ \xi &= \eta \mu + \sum_{k=1}^q \omega_k v_k, & m &= \mu^\top Q \xi, & \psi_k &= v_k^\top Q \xi, \\ \sigma^2 &= \frac{\gamma^2}{n} \text{tr}((Q \Sigma)^2), & r &= y m + \sigma \tilde{e} + \sum_{k=1}^q \psi_k e_k, \end{aligned} \quad (64)$$

where $\tilde{e} \sim \mathcal{N}(0, 1)$ is independent of $(y, e_1, \dots, e_q)$. The associated Gaussian equivalent for the estimator is

$$\tilde{\theta} = Q \left(\xi + \gamma \Sigma^{1/2} u \right), \quad u \sim \mathcal{N}(0, I_p/n). \quad (65)$$

Thus its mean is $Q \xi$ and its covariance is $\gamma^2 Q \Sigma Q / n$.

We now recover these quantities from Theorem 4.4. For ridge regularization, set

$$U_* := x^\top \mu_* + \alpha_* z, \quad h_* := h_{\kappa_*}(U_*, y).$$

The definitions (12) and (63) give $\xi_y(U_*, \kappa_*) = -\kappa_* h_*$. Gaussian integration by parts in z therefore transforms the ν -equation into

$$\nu_* = -\frac{1}{\alpha_*} \mathbb{E}[z h_*] = -\mathbb{E}[\partial_u h_{\kappa_*}(U_*, y)]. \quad (66)$$

Hence Mai–Liao’s ϑ is our ν_* . Likewise, with $\gamma^2 := \mathbb{E}[h_*^2]$, the α -equation becomes

$$\alpha_*^2 = \frac{\gamma^2}{n} \text{tr}((Q \Sigma)^2) = \sigma^2, \quad (67)$$

after the fixed- q LFMM reduction. Here $C_x - \Sigma$ has rank $O(q)$, so it does not affect the normalized trace equations for κ and α ; the informative finite-rank part is retained explicitly in the vector equation.

Indeed, the vector fixed-point equation reads

$$\lambda \mu_* = \mathbb{E}[h_* x]. \quad (68)$$

Adding $\nu_* \Sigma \mu_*$ to both sides and setting $Q = (\lambda I_p + \nu_* \Sigma)^{-1}$ gives

$$\mu_* = Q (\mathbb{E}[h_* x] + \nu_* \Sigma \mu_*). \quad (69)$$

In the LFMM reduction, the q informative factors are kept explicitly, whereas integration by parts (or the corresponding universality argument) cancels the uninformative terms through $\mathbb{E}[e_k h_*] + \nu_* v_k^\top \mu_* = 0$ for $k > q$. Defining

$$\begin{aligned}\eta &:= \mathbb{E}[y h_*], \\ \omega_k &:= \mathbb{E}[e_k h_*] + \nu_* v_k^\top \mu_*, \quad k \leq q, \\ \xi &:= \eta \mu + \sum_{k=1}^q \omega_k v_k,\end{aligned}$$

equation (69) becomes

$$\mu_* = Q\xi.$$

Consequently, $m = \mu^\top \mu_* = \mu^\top Q\xi$ and $\psi_k = v_k^\top \mu_* = v_k^\top Q\xi$. Together with (66)–(67), the scalar score U_* reduces to

$$U_* \stackrel{d}{=} ym + \sigma \tilde{e} + \sum_{k=1}^q \psi_k e_k,$$

which is precisely the scalar variable r in (64). Finally, (65) has mean $Q\xi = \mu_*$ and produces the same covariance-weighted fluctuation scale as (67). This identifies the Mai–Liao equations as the LFMM specialization of our fixed-point system, while keeping explicit the fixed- q reduction used in their analysis.

G. Failure of Universality on MNIST Data

MNIST-based data model. We consider a binary classification task constructed from MNIST by merging digits $\{3, 6\}$ into class 0 and digits $\{4, 7\}$ into class 1. Samples are represented either by raw pixels, and we compare the original data to a Gaussian surrogate matched in mean and covariance.

Left panel: failure of score universality. The left panel of Figure 2 shows the empirical distribution of the decision score $\hat{\theta}^\top x$. We compare it to a Gaussian distribution with the same empirical mean and variance, as commonly assumed, as well as to our theoretical prediction derived in the non-Gaussian setting. While the Gaussian approximation fails to capture the empirical histogram, our theoretical density closely matches the observed distribution, demonstrating that the score is distinctly non-Gaussian despite moment matching.

Right panel: consequences for performance. The right panel illustrates the impact of this non-Gaussianity on performance predictions. Performance estimates obtained under the Gaussian score assumption (green) exhibit a significant mismatch with empirical results. In contrast, our corrected theoretical predictions accurately track the observed performance. Additionally, we report the performance obtained when the original data are replaced by a Gaussian distribution with matching moments (purple), which also differs substantially from the empirical MNIST performance. This shows that matching first and second moments is insufficient to guarantee universality at the level of generalization performance.

Together, these results demonstrate that both score universality and Gaussian replacement universality can fail in realistic, structured datasets such as MNIST, reinforcing the need for non-Gaussian theoretical corrections.